AI System for Sinus CT Analysis

Graduation Project 2 prepared to obtain the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science

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SUPERVISOR CERTIFICATION

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AI System for Sinus CT Analysis

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was carried out under my supervision at the Faculty of Artificial Intelligence Engineering in partial fulfillment of the requirements for the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science.

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كلمة شكر وتقدير

الحمد لله رب العالمين، الذي وفقني وأعانني على إتمام هذا العمل، وما كان لهذا الجهد أن يكتمل لولا فضل الله أولاً، ثم دعم وتوجيه أشخاص كان لهم أثر كبير في هذه المسيرة العلمية.

أتقدم بأسمى آيات الشكر والتقدير إلى **المهندس أنس عبد العزيز**، الذي كان له الدور الأكبر في مرافقتي ودعمي خلال ثلاث سنوات من العمل والتعلّم. فقد كان موجهاً صبوراً، وداعماً حقيقياً، لم يبخل بعلمه ولا بوقته ولا بنصحه، وكان حاضراً في المراحل الصعبة قبل السهلة. لقد علّمني أن النجاح لا يأتي دفعة واحدة، بل يبنى بالصبر، والمثابرة، والإصرار، وأن العمل العلمي يحتاج إلى دقة ومسؤولية وإيمان بالفكرة. فله مني كل الامتنان والتقدير على تحمّله، وتشجيعه، وثقته، وعلى كل ما قدمه من علم وخبرة وتوجيه كان له أثر واضح في إنجاز هذا المشروع.

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الملخص

يعالج هذا المشروع مشكلة تحليل صور التصوير الطبقي المحوري للجيوب الأنفية باستخدام تقنيات التعلم العميق. وتكمن أهمية هذه المسألة في أن صور الجيوب الأنفية تحتوي على بُنى تشريحية متداخلة، كما أن تقييمها يدوياً قد يكون مرهقاً ويتطلب خبرة اختصاصية. وتظهر تحديات إضافية بسبب محدودية البيانات، وعدم توازن الحالات المرضية، واحتمال ترافق أكثر من شذوذ تشريحي أو مرضي في الحالة الواحدة.

يقترح المشروع خط معالجة يعتمد أولاً على التقسيم التشريحي باستخدام نموذج SwinUNETR ، وذلك لتحديد مناطق مهمة مثل الجيب الفكي الأيمن، والجيب الفكي الأيسر، والتجويف الأنفي الأيمن، والتجويف الأنفي الأيسر، والبلعوم الأنفي. بعد ذلك، يتم استخراج مناطق اهتمام موجّهة تشريحياً، ثم تدريب نموذج تصنيف متعدد التسميات للكشف عن خمس علامات مرضية مختارة. وقد تم اعتماد نموذج ConvNeXt-Tiny كنموذج التصنيف النهائي بعد مقارنته مع نموذج Swin-T.

أظهرت نتائج التقسيم قدرة جيدة على تحديد البُنى التشريحية، حيث بلغ متوسط معامل Dice النهائي 87.43%، وبلغ متوسط IoU مقدار 81.02% على مجموعة الاختبار. أما في مرحلة التصنيف، فقد حقق نموذج ConvNeXt-Tiny أداءً جيد وفقاً لمتوسط Macro F1 عبر الطيات حيث بلغ 78.49% ونتائج التنبؤ خارج الطية OOF F1 69.28% . ومع ذلك، بقيت بعض التسميات، وخاصة شذوذ الجيب الفكي، أقل استقراراً بسبب انخفاض عدد العينات الإيجابية وترافقها المتكرر مع شذوذات أخرى.

Abstract

This project addresses the problem of artificial-intelligence-assisted analysis of sinus computed tomography (CT) scans. Sinus CT interpretation is clinically important because it supports the assessment of sinonasal anatomy and pathology, but automatic analysis is challenging due to the three-dimensional nature of CT volumes, anatomical variability, limited annotated data, and strong class imbalance among abnormal findings.

The proposed solution follows a segmentation-assisted pipeline. First, a SwinUNETR model is trained to segment key anatomical structures in NasalSeg CT volumes. Second, anatomy-guided regions of interest (ROIs) are extracted from the CT scan and segmentation masks. Third, a multi-label classifier predicts five final abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. ConvNeXt-Tiny is selected as the final classifier after comparison with Swin-T.

The segmentation model achieved a mean Dice score of 0.8743 and mean IoU of 0.8102 on the test set. For classification, ConvNeXt-Tiny achieved a fold-level macro F1 score of 0.7849 and an out-of-fold macro F1 score of 0.6928, outperforming Swin-T in the final comparison. The results show that the proposed pipeline is promising, while also revealing important limitations caused by small positive sample counts, multi-label co-occurrence, and indirect anatomical targets such as turbinate abnormalities.

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Glossary of Terms and Symbols

Table 0.1: Glossary of terms and abbreviations

Term	English Equivalent / Description	Symbol / Abbreviation
Computed Tomography	Cross-sectional X-ray imaging modality used to visualize internal anatomy.	CT
Region of Interest	A selected image region used as model input for a specific label.	ROI
Dice Similarity Coefficient	Overlap metric used to evaluate segmentation masks.	Dice
Intersection over Union	Overlap metric between predicted and reference masks.	IoU
Convolutional Neural Network	Deep learning architecture based on convolutional operations.	CNN
Vision Transformer	Neural architecture that applies self-attention to image patches.	ViT
SwinUNETR	Transformer-based 3D U-shaped segmentation architecture.	SwinUNETR
ConvNeXt	Modern convolutional architecture inspired by transformer-era design choices.	ConvNeXt
Out-of-Fold Prediction	Prediction produced for each sample by a model that did not train on it.	OOF
Focal Loss	Classification loss that reduces the contribution of easy examples.	Focal Loss
Average Precision	Area under the precision-recall curve, useful for imbalanced classification.	AP
Exact Match Accuracy	Multi-label metric requiring all labels of a case to be predicted correctly.	Subset Accuracy

Chapter One: Introduction

1.1 Introduction

Computed tomography (CT) is one of the most important imaging modalities for evaluating the paranasal sinuses and nasal cavity. It provides detailed cross-sectional views of bony and soft-tissue structures, making it useful for diagnosing sinus disease, studying anatomical variation, and planning endoscopic sinus surgery. However, CT volumes are three-dimensional and usually contain many slices, which makes manual review time-consuming and dependent on clinical experience.

Artificial intelligence, especially deep learning, has become an important research direction in medical image analysis. Deep learning models can learn visual patterns directly from imaging data and can be trained to perform segmentation, classification, detection, and quantification tasks. In sinus CT analysis, segmentation can localize anatomical structures, while classification can support the detection of abnormal findings. Combining these two tasks may help produce a system that is more anatomically meaningful than a purely image-level classifier.

This project focuses on developing an AI system for sinus CT analysis using a segmentation-assisted classification pipeline. The system first localizes important sinonasal structures using a SwinUNETR segmentation model, then extracts anatomy-guided regions of interest, and finally classifies five selected abnormality labels using modern image-classification backbones. The final classification model selected in this work is ConvNeXt-Tiny because it achieved the strongest performance among the final evaluated classifiers.

1.2 Scientific Problem of the Research Project

The scientific problem addressed in this project is the automatic analysis of sinus CT scans for selected anatomical and pathological abnormalities. The problem is not a simple image classification task because the abnormalities are region-dependent, multi-label, and often correlated. A patient may have septal deviation, turbinate hypertrophy, and maxillary sinus abnormality at the same time. A single-label classifier or a simple dominant-label confusion matrix cannot fully represent this situation.

The problem also includes data-related challenges. The dataset contains a limited number of CT cases, and the positive counts differ strongly across labels. For example, the final labels include 71 positive cases for nasal septum abnormality, 60 positive cases for each inferior turbinate hypertrophy label, but only 22 and 23 positives for the right and left maxillary abnormality labels. This imbalance makes maxillary abnormalities less stable during training and evaluation.

Another difficulty is the mismatch between available segmentation masks and some clinical targets. NasalSeg provides masks for the right and left maxillary sinuses, right and left nasal cavities, and nasopharynx. It does not directly segment the nasal septum or turbinates. Therefore, the final model must infer septal and turbinate-related abnormalities using contextual neighboring regions rather than direct target masks.

1.3 Research Objectives

The main objective of this project is to develop a segmentation-assisted AI system for sinus CT analysis that localizes relevant sinonasal anatomy and predicts selected abnormality labels using anatomy-guided regions of interest.

The secondary objectives are:

- Train a 3D SwinUNETR segmentation model to localize the available NasalSeg anatomical structures: right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx.
- Convert the clinical annotation workbook into a model-ready multi-label format for five selected abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy.
- Design anatomy-guided ROI extraction rules that connect each abnormality label to the most relevant available segmentation region.
- Handle missing direct masks for the nasal septum and inferior turbinates using medically justified contextual ROIs.
- Evaluate ConvNeXt-Tiny and Swin-T as classification backbones under the same 5-fold out-of-fold evaluation setting.
- Select the best-performing final classifier and analyze its strengths, weaknesses, and limitations, especially under class imbalance and label co-occurrence.

1.4 Research Contribution

The main contribution of this project is the development of a segmentation-assisted multi-label classification workflow for sinus CT analysis. Instead of classifying the full CT volume without localization, the proposed system first segments relevant sinonasal structures and then uses these masks to extract anatomy-guided ROIs. This links each abnormality prediction to a clinically meaningful region and makes the pipeline more interpretable than a blind full-volume classifier.

A second contribution is the final ROI design. Because NasalSeg does not directly provide septum or turbinate masks, the project explores contextual regions based on the available nasal cavity masks. The final septum ROI is defined using both nasal cavities with padding, representing the idea that the septum lies between the right and left nasal cavity regions. The inferior turbinate labels are modeled using the ipsilateral nasal cavity ROIs.

A third contribution is the comparative evaluation of ConvNeXt-Tiny and Swin-T for this specific multi-label sinus CT task. ConvNeXt-Tiny is selected as the final model because it achieves better fold-level and out-of-fold performance than Swin-T. The report also discusses why some labels, especially maxillary abnormalities, remain unstable due to low positive counts and label co-occurrence.

The contribution is therefore not only the use of ConvNeXt-Tiny as a classifier, but the complete medical-AI workflow that connects sinus CT segmentation, ROI generation, multi-label abnormality prediction, and careful evaluation under limited-data conditions.

1.5 Target Beneficiaries

- Radiologists and ENT physicians who interpret sinus CT scans.
- Hospitals and imaging centers interested in AI-assisted triage or decision support.
- Researchers working on medical image segmentation and classification.
- Students and engineers studying applied deep learning in medical imaging.
- Developers who may extend the system into a clinical review or educational tool.

1.6 Report Organization

Chapter Two presents the theoretical background needed to understand the project, including CT imaging, sinus anatomy, segmentation, classification, CNNs, transformers, and evaluation metrics. Chapter Three reviews related work and identifies the research gap. Chapter Four explains the research methodology, dataset, preprocessing, segmentation model, ROI extraction, and final classification models. Chapter Five presents the experimental results and discusses the segmentation and classification outcomes. Chapter Six concludes the report and proposes future work.

Chapter Two: Theoretical Background

2.1 Computed Tomography and Sinus Imaging

Computed tomography constructs cross-sectional images from X-ray measurements acquired around the patient. In sinus imaging, CT is especially valuable because it shows the bony boundaries of the paranasal sinuses and nasal cavity. The intensity values in CT images are commonly represented in Hounsfield Units (HU), where air has very low values and bone has high values. This contrast is useful for distinguishing air-filled cavities from surrounding structures.

Sinus CT volumes are usually analyzed in axial, coronal, and sagittal planes. A model trained on CT data must therefore learn three-dimensional spatial relationships, or approximate them through multi-view two-dimensional representations. In this project, segmentation is performed as a 3D task, while classification uses 2.5D ROI images derived from each anatomical crop.

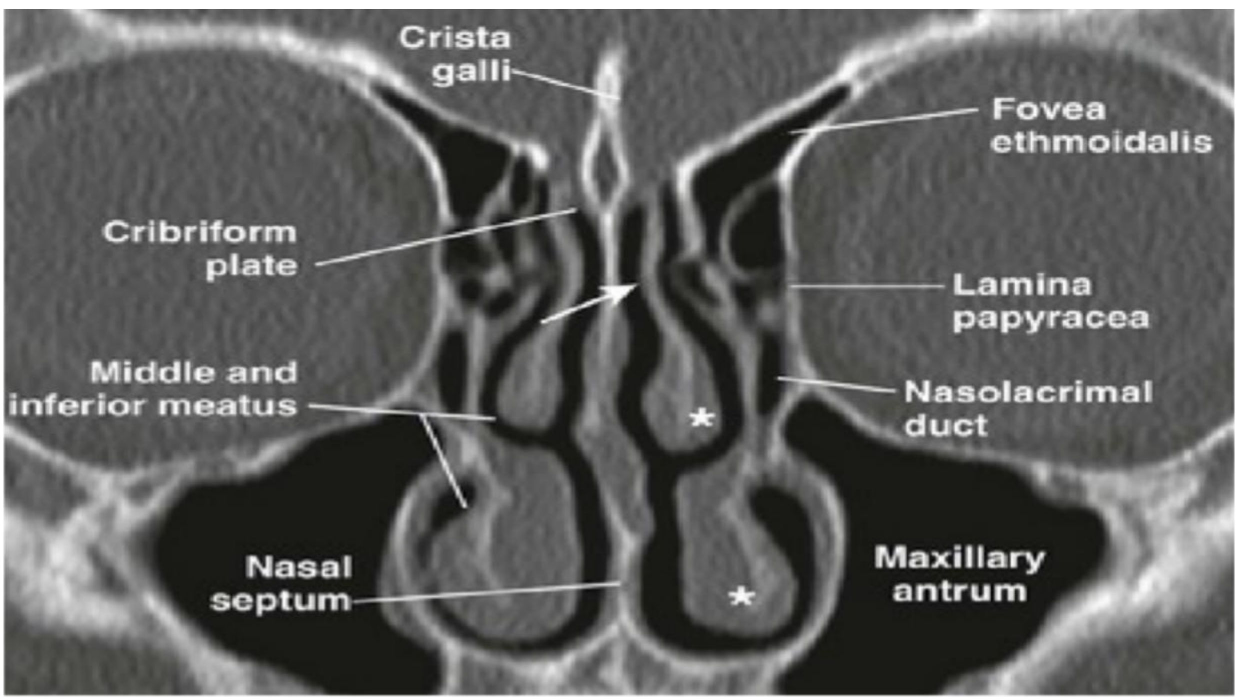

Figure 2.1: Main sinonasal anatomical regions in CT imaging

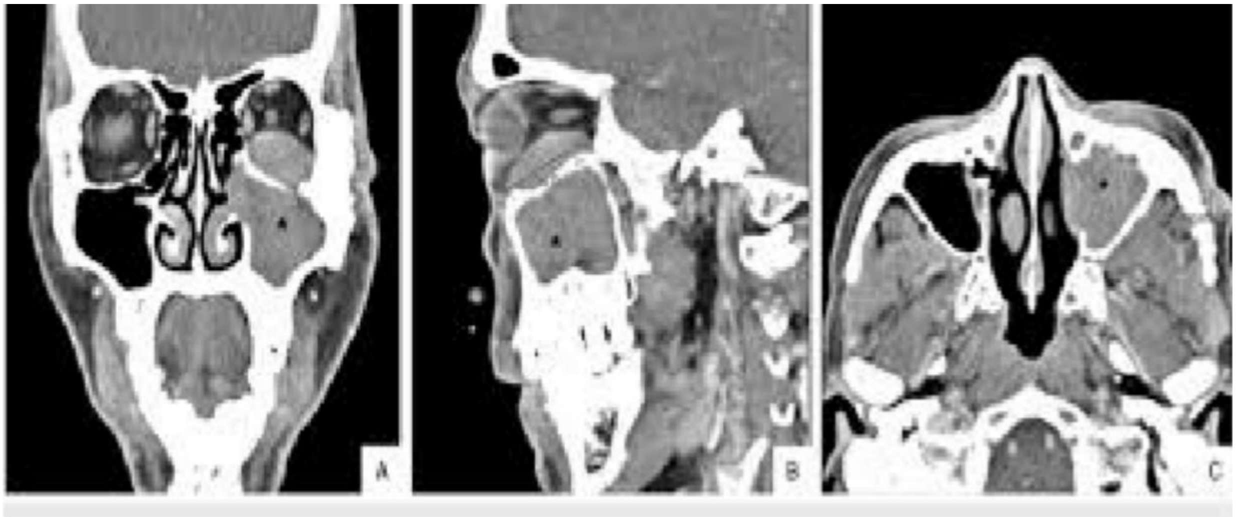

Figure 2.2: Example CT views of the sinonasal area, coronal, sagittal, axial

In this project, CT imaging is important not only as a diagnostic modality but also as the input source for both segmentation and classification. The air-filled sinus cavities, bony boundaries, and soft-tissue changes provide different intensity patterns that can be learned by the model. This is why CT windowing and multi-plane analysis are important for the proposed pipeline. The segmentation stage uses the 3D CT volume to localize anatomy, while the classification stage uses axial, coronal, and sagittal ROI views to preserve clinically useful context around each selected sinus region.

2.2 Sinonasal Anatomy Relevant to the Project

The paranasal sinuses are air-filled cavities surrounding the nasal cavity. In this project, the most important sinus structures are the right and left maxillary sinuses because they are directly related to the right and left maxillary abnormality labels. These regions are suitable for ROI-based analysis because the NasalSeg dataset provides separate segmentation masks for both maxillary sinuses.

The nasal cavity is divided into right and left sides by the nasal septum. Septal abnormality is clinically important because deviation or structural abnormality of the septum can affect nasal airflow and may coexist with other sinonasal findings. However, the dataset used in this project does not provide a direct nasal septum mask. Therefore, the septum label is handled using a contextual ROI derived from both nasal cavities, since the septum lies between them.

The inferior turbinates are anatomical structures located inside the nasal cavity and are involved in airflow regulation. Inferior turbinate hypertrophy can narrow the nasal airway and may appear together with septal or sinus abnormalities. Since direct turbinate masks are not available, the right and left inferior turbinate hypertrophy labels are modeled using the corresponding right and left nasal cavity ROIs.

Therefore, the anatomical background directly affects the design of the AI pipeline. Maxillary abnormality uses direct maxillary sinus ROIs, nasal septum abnormality uses a bilateral nasal cavity context ROI, and inferior turbinate hypertrophy uses ipsilateral nasal cavity ROIs. This connection between anatomy and ROI design is one of the main reasons the proposed system is more interpretable than a full-volume classifier.

2.3 Medical Image Segmentation

Medical image segmentation is the task of assigning a class label to each pixel or voxel. In 3D CT segmentation, each voxel in the volume is assigned to an anatomical class such as background, right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, or nasopharynx. Segmentation is useful because it localizes structures and enables measurement, visualization, and region-specific analysis.

In this project, segmentation is not the final clinical output. Instead, it is an intermediate stage used to guide ROI extraction. This design allows the classification model to focus on relevant anatomical regions rather than the entire CT volume.

$$\boxed{Dice = 2\,|P \cap G| \,/\, (|P| + |G|)}$$

Equation [2.1]: Dice Similarity Coefficient

$$IoU = |P \cap G| \,/\, |P \cup G|$$

Equation [2.2]: Intersection over Union

2.4 Deep Learning for CT Analysis

Deep learning models learn hierarchical representations from data. Early layers usually capture simple intensity and edge patterns, while deeper layers capture more complex anatomical or pathological features. For CT analysis, the model must be robust to anatomical variation, acquisition differences, and class imbalance.

There are two broad approaches to CT learning. A 3D model processes the full volume or volume crops and can preserve volumetric context, but it requires more memory and more data. A 2D or 2.5D model processes slices or multi-plane views, reducing memory requirements and enabling the use of pretrained natural-image backbones. This project uses 3D modeling for segmentation and 2.5D transfer learning for classification.

2.5 Convolutional Neural Networks and ConvNeXt

Convolutional neural networks are well suited for medical images because they exploit local spatial structure. ConvNeXt is a modern convolutional architecture that updates classical CNN design choices while preserving the simplicity and locality of convolutional operations. In this project, ConvNeXt-Tiny is used as the final classification backbone because it showed the best balance between performance and stability on the selected multi-label task.

The selection of ConvNeXt-Tiny is supported by two ideas. First, ConvNeXt preserves convolutional inductive bias, which is often helpful in limited-data medical imaging. Second, its design includes transformer-era improvements while avoiding some of the data demands of pure transformer backbones. This makes it a reasonable candidate for ROI-based 2.5D classification, where the input already focuses on a clinically meaningful region.

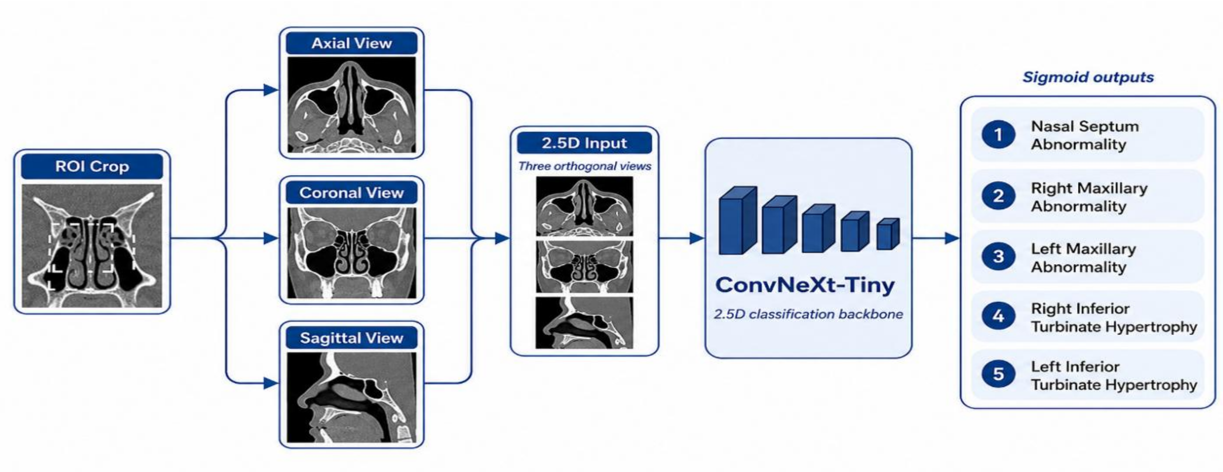

Figure 2.3: Simplified ConvNeXt-Tiny classification concept

2.6 Vision Transformers, Swin Transformer, and SwinUNETR

Vision transformers process images through patch representations and self-attention. Standard transformers can be computationally expensive for high-resolution medical images. Swin Transformer addresses this issue by using hierarchical shifted windows, which makes attention more efficient while still capturing contextual information. SwinUNETR combines a Swin Transformer encoder with a U-shaped segmentation decoder, making it suitable for 3D medical segmentation tasks.

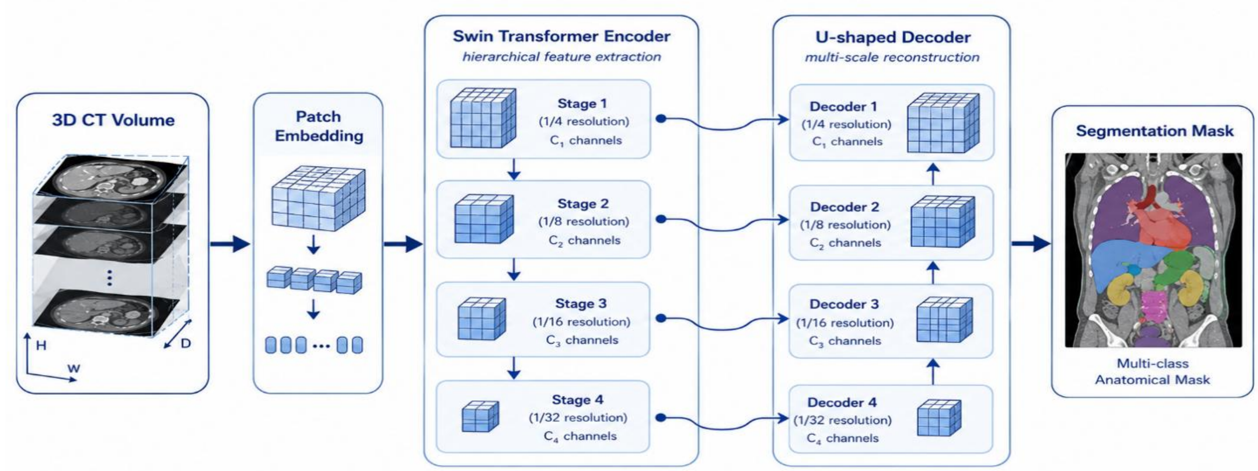

Figure 2.4: Simplified SwinUNETR architecture overview

2.7 Multi-Label Classification

In multi-label classification, each case can belong to more than one class. This is different from multi-class classification, where each sample belongs to exactly one class. The sinus CT task in this project is multi-label because one case may contain multiple simultaneous abnormalities. For this reason, each final label is predicted with an independent sigmoid output rather than a single softmax output.

$$\hat{y}_i = 1 \text{ if } \mathrm{sigmoid}(z_i) \geq \tau_i, \text{ otherwise } 0$$

Equation [2.3]: Binary classification decision from sigmoid probability

This formulation matches the clinical nature of sinus CT findings. A single patient may have septal abnormality, turbinate hypertrophy, and maxillary sinus abnormality in the same scan. Therefore, the model must not force the case into only one class. In this project, a normal case is represented by an all-zero vector across the five selected abnormality labels, while abnormal cases may have one or more positive labels.

2.8 Class Imbalance and Focal Loss

Class imbalance occurs when some labels have many positive examples while others have very few. This is an important issue in the proposed sinus CT classification task because the maxillary abnormality labels have fewer positive cases than the septum and inferior turbinate labels. As a result, the model may learn the more frequent labels more easily while producing unstable predictions for rare labels.

In medical AI, this problem is important because rare positive findings should not be ignored. A model that achieves high accuracy by predicting mostly negative labels may still be clinically weak. For this reason, the final classification experiments used focal loss with positive class weighting. Focal loss reduces the influence of easy examples and gives more attention to difficult or misclassified samples. Positive class weighting helps compensate for labels with fewer positive cases.

In this project, focal loss was used with gamma equal to 1. This setting was selected to handle imbalance without making the training overly aggressive for rare labels. The goal was not only to improve overall accuracy, but to produce more balanced behavior across the five abnormality labels.

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

Equation [2.4]: Binary focal loss concept

2.9 Evaluation Metrics

2.9.1 Segmentation Metrics

The segmentation model was evaluated with spatial overlap metrics and runtime measures. Dice and IoU measure agreement between predicted and reference masks. Precision measures the proportion of predicted foreground voxels that are correct, while recall or sensitivity measures the proportion of reference voxels captured by the model. Inference time and throughput describe the practical speed of the trained model during test-time prediction.

$$Precision = TP / (TP + FP)$$

Equation [2.5]: Precision

$$Recall = TP / (TP + FN)$$

Equation [2.6]: Recall / Sensitivity

$$Specificity = TN / (TN + FP)$$

Equation [2.7]: Specificity

2.9.2 Classification Metrics

The classification task was evaluated using complementary metrics because the task is imbalanced and multi-label. Accuracy measures label-wise correctness. Precision and recall measure false-positive and false-negative behavior. Specificity measures the ability to reject negative cases. Balanced accuracy averages sensitivity and specificity, making it more informative when positive and negative cases are imbalanced.

F1-score is the harmonic mean of precision and recall. Macro F1 averages label-wise F1 scores, giving each abnormality equal importance regardless of its frequency. AUC measures ranking quality across thresholds, while average precision summarizes the precision-recall curve and is useful for imbalanced labels. Overall label accuracy measures correctness over all labels, while exact match accuracy requires every label of a case to be predicted correctly.

The report also uses out-of-fold evaluation. In a 5-fold setting, each case is predicted by a model that did not train on it. This produces a more reliable estimate of generalization than evaluating only on the training folds. Threshold tuning is performed on the OOF predictions to select label-specific probability cutoffs.

For the final classification task, macro F1 is especially important because it gives equal importance to each abnormality label. This is more suitable than simple accuracy for the proposed dataset, where the positive counts are not balanced across labels. Out-of-fold evaluation is also important because it provides one validation prediction for every CT case using a model that did not train on that case. This makes the final classification result more reliable than depending only on a single train-test split. The 5×5 dominant-label confusion matrix is used only as a visualization because it compresses a multi-label prediction into one dominant true label and one dominant predicted label.

$$F1 = 2 \times Precision \times Recall / (Precision + Recall)$$

Equation [2.8]: F1-score

$$Balanced\ Accuracy = (Sensitivity + Specificity) / 2$$

Equation [2.9]: Balanced accuracy

$$Exact\ Match = number\ of\ cases\ with\ all\ labels\ correct / total\ number\ of\ cases$$

Equation [2.10]: Exact match accuracy

2.10 Chapter Summary

This chapter presented the theoretical background needed for the proposed sinus CT analysis system. It explained CT imaging, relevant sinonasal anatomy, medical image segmentation, 3D and 2.5D deep learning, ConvNeXt-Tiny classification, SwinUNETR segmentation, multi-label prediction, class imbalance, and evaluation metrics. The chapter also connected these concepts to the specific design of this project, where segmentation is used to extract anatomy-guided ROIs and ConvNeXt-Tiny predicts five sinus-related abnormality labels.

Chapter Three: Literature Review

3.1 Introduction

This chapter reviews previous studies related to sinus CT segmentation, paranasal anomaly classification, medical-image backbones, and 2.5D transfer learning. Similar to the organization used in the ChromoSep-AI literature review, the chapter discusses the most relevant studies separately rather than reducing them to a single table. This structure makes it clearer how each work influenced the design of the proposed segmentation-assisted sinus CT analysis system.

The reviewed literature shows that the field contains strong segmentation studies and strong general classification backbones, but fewer works combine anatomical sinus segmentation with multi-label abnormality prediction. Therefore, the review focuses on the research gap between segmentation-only systems and abnormality-classification systems.

3.2 Previous Studies in Sinus CT Segmentation

Previous sinus CT segmentation studies mainly focused on localizing anatomical structures such as the maxillary sinuses, nasal cavity, nasopharynx, and other paranasal sinus regions. These works are important for this project because anatomical localization is the first stage of the proposed pipeline. Instead of using segmentation as the final clinical output, this project uses segmentation to guide ROI extraction before abnormality classification. The reviewed segmentation papers therefore provide the technical foundation for using 3D models such as SwinUNETR and U-Net variants in sinonasal CT analysis.

3.3 NasalSeg Dataset and Benchmark

Zhang et al. introduced NasalSeg, an open-access dataset for nasal cavity and paranasal sinus segmentation from 3D CT images [1]. The dataset contains 130 CT scans with voxel-wise annotation of five structures: right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. This work is the most directly related foundation for the present project because the proposed system uses the same anatomical structures to guide ROI extraction. However, NasalSeg does not directly annotate the nasal septum or inferior turbinates, which creates an important limitation for downstream abnormality classification.

3.4 SwinUNETR for 3D Medical Image Segmentation

Hatamizadeh et al. proposed SwinUNETR, a transformer-based encoder-decoder architecture for 3D medical image segmentation [3]. The model uses a Swin Transformer encoder to capture hierarchical volumetric context and a U-shaped decoder to reconstruct dense segmentation masks. In their medical segmentation benchmarks, transformer-based 3D modeling showed strong Dice performance and motivated its use for volumetric anatomical localization. In this project, SwinUNETR was selected for the segmentation stage because sinus CT volumes require spatial context across slices.

3.5 Whangbo et al. Paranasal Sinus Segmentation

Whangbo et al. compared several 3D U-Net variants for multi-class paranasal sinus segmentation in sinusitis CT images [6]. The normal 3D U-Net achieved the best reported F1 score, reaching 84.29% on the normal test set and 79.32% on the abnormal test set. This study supports the use of 3D CNN-based segmentation for sinus anatomy, especially when sinusitis changes the appearance of the target regions. Its limitation for the present project is that it focuses on anatomical segmentation rather than multi-label abnormality prediction.

3.6 Ozturk et al. Maxillary Sinus Segmentation

Ozturk et al. developed a U-Net-based method for automatic maxillary sinus segmentation on cone-beam CT images [9]. The study is relevant because maxillary sinus abnormality is one of the final target groups in this project. It demonstrates that deep learning can localize maxillary sinus boundaries effectively, which is important for region-specific analysis. However, the work is CBCT-focused and does not address a full sinus CT pipeline with septum, turbinate, and bilateral multi-label abnormalities.

3.7 Song et al. CNN, Transformer, and Hybrid Segmentation Comparison

Song et al. compared CNN, vision-transformer, and hybrid networks for paranasal sinus segmentation in CT images [10]. Their results showed that Swin UNETR achieved the strongest overall segmentation performance among the tested models, with a mean Jaccard Index of 0.719, Dice Similarity Coefficient of 0.830, recall of 0.758, and HD95 of 10.529. This study is important because it directly compares architecture families in the sinus domain. It also supports the idea that hybrid 3D architectures are strong candidates for sinus segmentation, although it remains focused on segmentation rather than diagnosis.

3.8 Previous Studies in Sinus and Paranasal Abnormality Classification

Previous classification studies in sinus and paranasal imaging have commonly focused on narrower diagnostic targets, especially maxillary sinus anomaly detection. These studies show that deep learning can support abnormality recognition in CT or CBCT images, but most of them do not address a five-label multi-label setting. In contrast, the current project predicts septum abnormality, bilateral maxillary abnormality, and bilateral inferior turbinate hypertrophy from the same CT case. This makes the classification problem more difficult because labels may co-occur and some target structures are not directly segmented.

3.9 Bhattacharya et al. Self-Supervised Maxillary Anomaly Classification

Bhattacharya et al. proposed a self-supervised learning approach for classifying paranasal anomalies in the maxillary sinus [7]. The method used a 3D convolutional autoencoder and a localization-oriented self-supervised task before fine-tuning for normal-versus-anomalous maxillary sinus classification. When trained with only 10% of the annotated dataset, the proposed method achieved an AUPRC of 0.79, outperforming BYOL, SimSiam, SimCLR, and SparK baselines. This is relevant to the present project because it addresses limited annotation, but it remains narrower because it focuses on maxillary sinus anomalies rather than five-label sinus CT classification.

3.10 Supervised Contrastive Learning for Maxillary Anomalies

Bhattacharya et al. also studied supervised contrastive learning for classifying paranasal anomalies in the maxillary sinus [8]. Their 3D CNN trained with supervised contrastive learning and cross-entropy achieved an AUROC of 0.85, compared with 0.66 for a 3D CNN trained with cross-entropy alone. This result shows that representation learning can improve classification in limited-data paranasal anomaly settings. However, the task was still focused on maxillary sinus anomaly classification and did not solve the broader problem of multi-label sinus CT abnormality prediction.

3.11 Classification Backbones and Transfer Learning Studies

The classification stage of this project is also connected to broader backbone and transfer-learning studies. Modern CNNs such as ConvNeXt and hierarchical transformers such as Swin

Transformer provide strong pretrained feature extractors that can be adapted to medical images. Because the available dataset is small, transfer learning is more practical than training a large classifier from scratch. The reviewed backbone studies justify the final comparison between ConvNeXt-Tiny and Swin-T and support the use of 2.5D ROI images as inputs to pretrained 2D models.

3.12 ConvNeXt

Liu et al. introduced ConvNeXt as a modernized convolutional architecture that adopts several transformer-era design choices while preserving the inductive bias of CNNs [4]. The ConvNeXt family achieved competitive large-scale vision performance, with the overall family reaching 87.8% ImageNet top-1 accuracy in the original work. This is relevant to the present project because the final classifier uses localized 2.5D ROI images and limited medical data, where convolutional locality can be beneficial. ConvNeXt-Tiny was therefore selected as the final backbone after outperforming the transformer comparison in the project experiments.

3.13 Swin Transformer

Liu et al. proposed Swin Transformer, a hierarchical vision transformer based on shifted windows [5]. The model achieved 87.3% ImageNet-1K top-1 accuracy and strong dense-prediction results, including 53.5 mIoU on ADE20K validation in the original paper. Swin is relevant because it represents the transformer family and motivates SwinUNETR for segmentation and Swin-T as a classification comparison. In this project, Swin-T is treated as an important comparison experiment rather than the final selected classifier.

3.14 MedNeXt and ConvNeXt-Style Medical Networks

Roy et al. introduced MedNeXt, a ConvNeXt-inspired architecture for medical image segmentation [12]. The model adapts large-kernel convolutional design to 3D encoder-decoder segmentation and shows that modern CNN principles can remain competitive in medical imaging. This supports the broader idea that convolutional models should not be dismissed in favor of transformers only. For the present project, this is conceptually important because ConvNeXt-Tiny achieved the best classification behavior on the limited ROI-based dataset.

3.15 Improved ConvNeXt-Tiny and CT Classification Studies

Recent medical classification studies have also adapted ConvNeXt-Tiny for efficient medical-image recognition [11]. FocalNeXt further shows that ConvNeXt-related components can be integrated into CT classification systems [13]. These studies are not sinus-specific, but they support the use of lightweight modern convolutional backbones in limited medical datasets. This literature is consistent with the final project result, where ConvNeXt-Tiny gave better fold-level and out-of-fold performance than Swin-T.

3.16 2.5D Medical Imaging Approaches

Several medical-imaging studies use 2.5D representations to balance volumetric context and computational efficiency [15]. In this approach, representative slices or orthogonal views are used as multi-channel inputs for 2D pretrained networks. This strategy is relevant to the current project because each 3D ROI is converted into axial, coronal, and sagittal views. The resulting representation preserves partial 3D context while enabling transfer learning with pretrained ConvNeXt-Tiny and Swin-T backbones.

3.17 State-of-the-Art Frameworks Analysis

The current state of the art in sinus CT analysis can be understood as two related directions. The first direction is high-quality anatomical segmentation, where 3D CNNs, SwinUNETR, and hybrid transformer-CNN models achieve strong Dice or Jaccard scores. The second direction is abnormality classification, where self-supervised, contrastive, and transfer-learning methods help under limited annotations. The proposed project is positioned between these two directions by using segmentation as an anatomical guide for multi-label abnormality prediction.

This state-of-the-art framing fits the project because the system does not claim to replace radiological interpretation. Instead, it combines a validated segmentation-style workflow with localized ROI classification. This makes the model more interpretable than a full-volume classifier and more clinically relevant than a segmentation-only system.

For the proposed project, the most relevant state-of-the-art direction is not only achieving high segmentation accuracy or high classification accuracy separately, but connecting both tasks into one medically meaningful workflow. The segmentation model localizes the anatomy, while the classifier uses this localization to focus on label-specific regions. This is important in sinus CT because abnormalities are not randomly distributed across the scan; they are associated with specific anatomical structures.

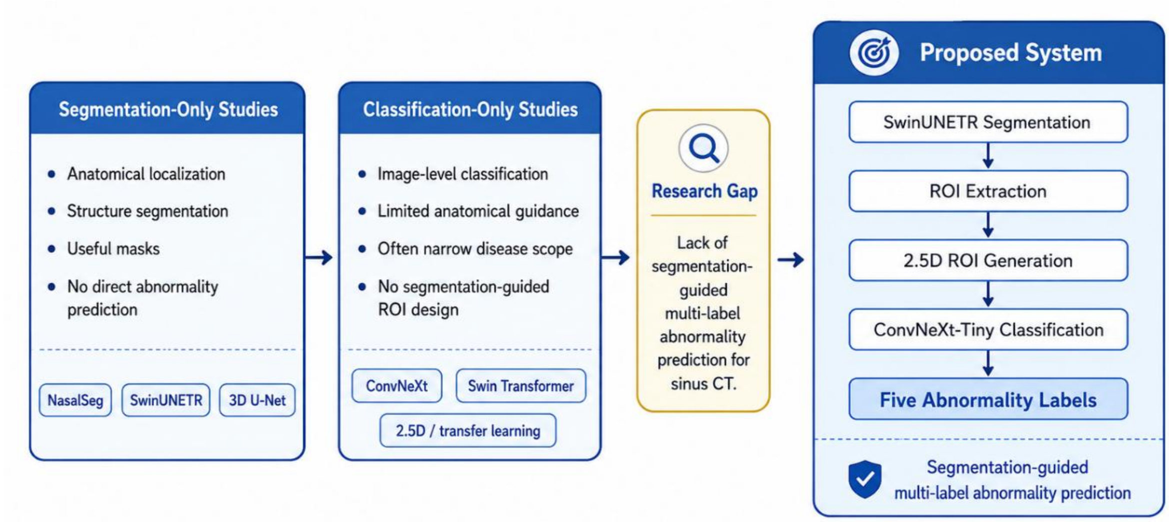

Figure 3.1: Research gap and positioning of the proposed system

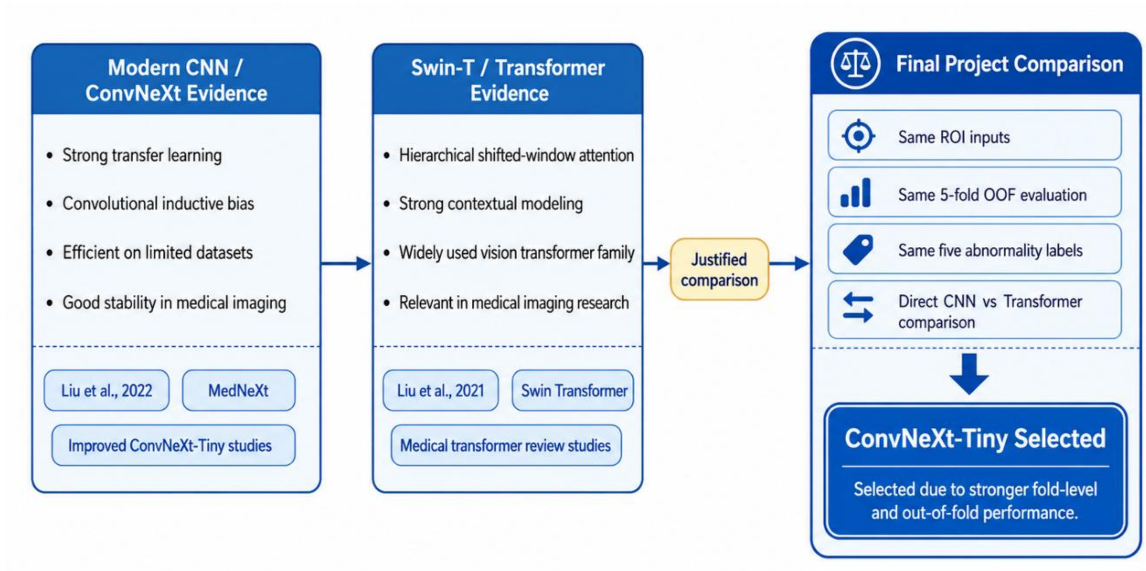

Figure 3.2: Literature-based justification of ConvNeXt-Tiny and Swin-T comparison

Table 3.1: Summary of related work in sinus CT analysis, backbones, and medical imaging

Paper	Approach	Architecture	Main	Strength	Limitation
Zhang et al., 2024, NasalSeg	Open-access CT dataset and benchmark for nasal cavity and paranasal sinus segmentation.	Dataset with voxel-level annotations; benchmark segmentation models.	Provides 130 3D CT scans with five annotated structures.	Public dataset directly related to this project.	Labels only five structures and do not directly include septum or turbinates.
	Transformer-	Swin		Strong	Memory

Hatamizadeh et al., 2022, SwinUNETR	based 3D medical image segmentation.	Transformer encoder with U-shaped decoder.	Captures long-range context in 3D segmentation.	architecture for volumetric medical segmentation.	intensive and originally validated on other medical tasks.
Liu et al., 2022, ConvNeXt	Modernized convolutional architecture.	Pure ConvNet with transformer-era design improvements.	Shows that modern CNNs can compete with transformer backbones.	Supports choosing ConvNeXt-Tiny for limited-data transfer learning.	Designed for natural images; medical transfer still requires careful ROI design.
Liu et al., 2021, Swin Transformer	Hierarchical transformer with shifted windows.	Window-based self-attention and multi-scale features.	Efficient transformer backbone for vision tasks.	Good comparison model against ConvNeXt.	Can require more data and careful regularization.
Whangbo et al., 2024	Multi-class paranasal sinus segmentation in sinusitis CT.	3D U-Net variants.	Segments major sinus regions in CT images.	Clinically relevant paranasal sinus segmentation setting.	Focuses on segmentation rather than abnormality prediction.
Bhattacharya et al., 2024	Self-supervised learning for paranasal anomaly classification.	3D convolutional autoencoder / self-supervised pipeline.	Classifies normal versus anomalous maxillary sinus patterns.	Addresses limited labeled data through self-supervision.	Narrower classification scope than multi-label abnormality prediction.
Ozturk et al., 2024	Automatic maxillary sinus segmentation on CBCT.	U-Net segmentation model.	Segments maxillary sinus structures.	Relevant to maxillary anatomy and dental/sinus imaging.	CBCT-focused and not a full multi-label CT abnormality pipeline.
Song et al., 2025	Comparison of CNNs, ViTs, and hybrid networks for sinus segmentation.	CNN, ViT, and hybrid segmentation networks.	Compares architecture families for paranasal sinus segmentation.	Useful for understanding architecture trade-offs.	Segmentation-focused and does not solve clinical multi-label classification.

Xia et al., 2025	Medical image classification using an improved ConvNeXt-Tiny.	Lightweight ConvNeXt-Tiny with pooling and attention refinements.	Shows ConvNeXt-Tiny can be adapted to medical classification.	Supports ConvNeXt-Tiny as a reasonable classification backbone.	Not specific to sinus CT or multi-label ROI classification.
Roy et al., 2023, MedNeXt	Medical image architecture based on ConvNeXt-style blocks.	3D ConvNeXt encoder-decoder.	Adapts ConvNeXt principles to medical imaging.	Supports modern CNNs in data-scarce medical settings.	Segmentation-focused, not a direct classifier for sinus CT.
Gulsoy and Kablan, 2025, FocalNeXt	CT lung cancer classification using ConvNeXt-augmented architecture.	ConvNeXt + FocalNet hybrid.	Applies ConvNeXt-related features to CT classification.	Shows ConvNeXt components are useful in CT classification.	Different disease and dataset.
Aburass et al., 2025	Review of vision transformers in medical imaging.	Review of ViT/Swin-style methods.	Summarizes transformer applications in medical classification and segmentation.	Justifies including Swin-T as a comparison.	Review-level evidence; not a sinus-specific experiment.
Wang et al., 2024	2.5D deep learning and conventional features for CT-based prediction.	2.5D masks/features plus model fusion.	Uses 2.5D representation to balance context and efficiency.	Supports 2.5D design for limited data.	Different clinical target and not sinus CT.

3.18 Comparative Analysis of Existing Systems

Existing work related to this project can be divided into three main groups: sinus CT segmentation studies, maxillary sinus abnormality classification studies, and general medical-image backbone studies. Direct comparison is difficult because these studies do not solve exactly the same task. Some focus only on anatomical segmentation, some classify only maxillary sinus anomalies, and others introduce general architectures rather than sinus-specific systems. However, they are still useful for understanding how the proposed system differs from previous approaches.

Table 3.2: Comparison of existing approaches and the proposed system

Study / Approach	Main task	Model / Method	Reported result / focus	Strength	Limitation compared with this project
Zhang et al., NasalSeg [1]	Nasal cavity and paranasal sinus segmentation	Dataset and benchmark segmentation models	130 CT scans with five annotated sinonasal structures	Directly related dataset and anatomical labels	Does not provide abnormality classification or direct septum/turbinate masks
Whangbo et al. [6]	Multi-class paranasal sinus segmentation	3D U-Net variants	Best reported F1 of 84.29% on normal test set and 79.32% on abnormal test set	Clinically relevant sinusitis CT segmentation	Segmentation-only; no multi-label abnormality prediction
Song et al. [10]	Paranasal sinus segmentation comparison	CNN, ViT, and hybrid models	Swin UNETR achieved Dice 0.830 and Jaccard 0.719	Shows value of hybrid/transformer segmentation in sinus CT	Does not address classification of abnormalities
Ozturk et al. [9]	Maxillary sinus segmentation	U-Net on CBCT	Automatic maxillary sinus localization	Relevant to maxillary sinus anatomy	CBCT-focused and not a full sinus CT multi-label system
Bhattacharya et al. [7]	Maxillary anomaly classification	Self-supervised 3D convolutional model	AUPRC 0.79 using 10% annotated data	Handles limited annotation for anomaly classification	Focuses only on maxillary anomaly, not five-label sinus CT classification
Bhattacharya et al. [8]	Maxillary anomaly classification	Supervised contrastive 3D CNN	AUROC 0.85 compared with 0.66 for cross-entropy baseline	Improves representation learning for limited data	Narrower task than the proposed multi-label setting
Proposed system	Segmentation-guided sinus CT abnormality classification	SwinUNETR + ROI extraction + ConvNeXt-Tiny	Segmentation Dice 0.8743; ConvNeXt-Tiny OOF macro F1 0.6928	Combines anatomical localization with five-label abnormality prediction	Limited by 130 cases, class imbalance, and missing direct septum/turbinate masks

3.19 Research Gap and Proposed Innovation

The reviewed studies show that sinus CT analysis has been studied mainly from two separate directions. The first direction focuses on anatomical segmentation, where models such as U-Net variants and SwinUNETR can localize sinus and nasal cavity structures. These studies are important, but they usually stop at segmentation and do not directly predict clinical abnormality labels.

The second direction focuses on abnormality classification, especially maxillary sinus anomaly detection. These studies show that deep learning can recognize abnormal findings in paranasal imaging, but most of them solve a narrower problem than the one addressed in this project. They usually classify maxillary sinus anomaly only, rather than predicting several co-existing abnormalities from the same CT case.

Therefore, the main research gap is the lack of a complete segmentation-assisted multi-label pipeline for sinus CT abnormality prediction. In the proposed problem, the model must predict five labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. This is more difficult than single-label classification because these findings may co-occur in one patient.

The proposed innovation is to connect anatomical segmentation with abnormality classification. SwinUNETR is used to localize available sinonasal structures, then these structures guide ROI extraction for ConvNeXt-Tiny classification. This makes the system more interpretable than a full-volume classifier because each prediction is linked to an anatomical region. The project also introduces contextual ROI handling for labels without direct masks: the septum is represented using both nasal cavities, and inferior turbinate hypertrophy is represented using the ipsilateral nasal cavity.

3.20 Chapter Summary

This chapter reviewed the main literature related to sinus CT segmentation, paranasal anomaly classification, modern CNN and transformer backbones, and 2.5D medical imaging. The review shows that the proposed project is positioned between segmentation-only sinus CT systems and image-level classification systems. Its main novelty is the use of segmentation-guided ROIs to support multi-label abnormality prediction while explicitly acknowledging the limitations caused by missing direct septum and turbinate masks.

Chapter Four: Research Methodology

4.1 Introduction

This chapter describes the complete methodology used to build the proposed AI system for sinus CT analysis. The system was designed as a segmentation-assisted multi-label classification pipeline rather than a direct full-volume classifier. This design was selected because the target abnormalities are anatomically localized: maxillary sinus abnormalities are related to the maxillary sinuses, nasal septum abnormality is related to the central region between the nasal cavities, and inferior turbinate hypertrophy is related to the nasal cavity regions.

The methodology consists of five main stages. First, CT volumes and anatomical masks are preprocessed to standardize orientation, spacing, intensity range, and tensor format. Second, a 3D SwinUNETR model is trained to segment the available NasalSeg structures. Third, the predicted or reference segmentation masks are used to extract anatomy-guided regions of interest. Fourth, each 3D ROI is converted into a 2.5D representation using axial, coronal, and sagittal views. Finally, a ConvNeXt-Tiny classifier predicts five abnormality labels using independent sigmoid outputs.

This design makes the classification stage more clinically meaningful than a blind image-level classifier. Instead of forcing the model to search the entire CT volume for every label, each prediction is connected to a specific anatomical region. This is especially important in sinus CT analysis because abnormalities may co-occur in the same patient, and some target structures, such as the nasal septum and inferior turbinates, are not directly segmented in the available dataset.

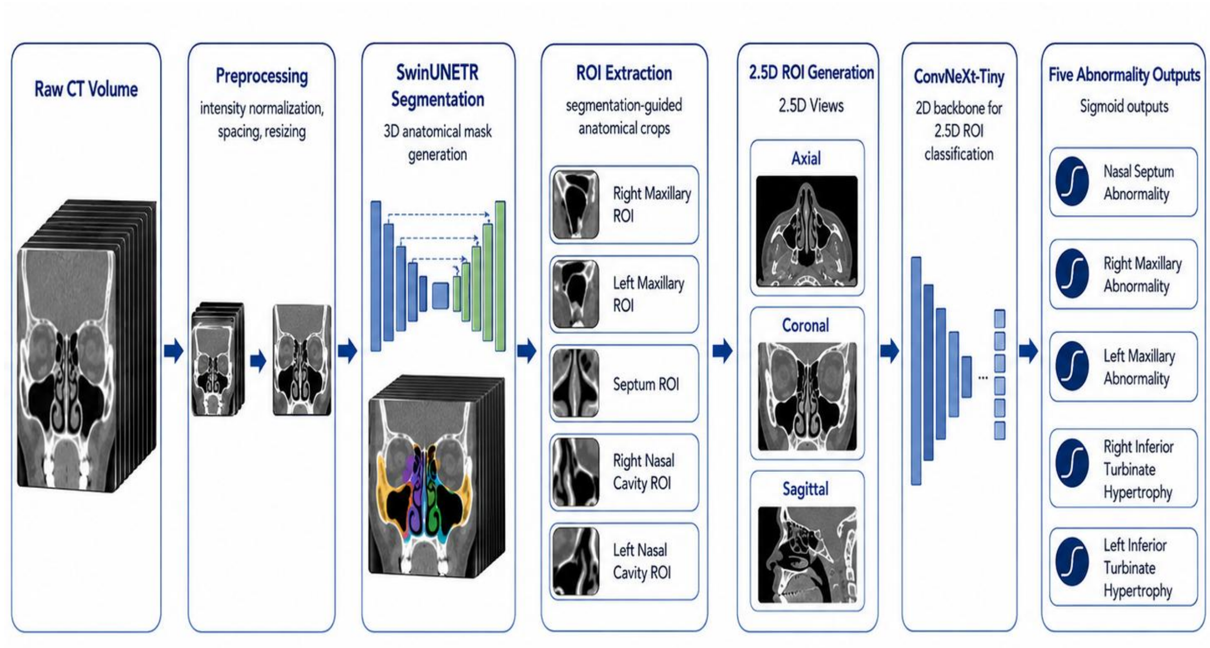

Figure 4.1: Overall pipeline of the proposed sinus CT analysis system

4.1.1 Classical and Baseline Approaches for Sinus CT Analysis

Before selecting the final deep-learning pipeline, several simpler approaches can be considered for sinus CT analysis. These approaches are useful as conceptual baselines, but they are limited for the specific objective of this project: segmentation-assisted multi-label abnormality prediction from sinus CT scans.

A traditional clinical approach is manual CT interpretation by a radiologist or ENT specialist. This remains the clinical reference standard because sinus CT findings require anatomical knowledge and clinical judgment. However, manual interpretation is time-consuming, depends on the reader's experience, and may vary between observers. This project does not aim to replace expert interpretation, but to build a research prototype that can assist anatomical localization and abnormality prediction.

Classical image-processing methods can also be considered for sinus CT because air-filled cavities and bony structures have different Hounsfield Unit ranges. Thresholding, region growing, connected-component analysis, and morphological operations may help isolate air cavities such as the maxillary sinuses. However, these methods are sensitive to anatomical variation, mucosal thickening, partial opacification, and irregular sinus boundaries. They are also not sufficient for predicting clinical abnormality labels such as septum abnormality or inferior turbinate hypertrophy.

Classical machine-learning methods, such as support vector machines or random forests, could be trained using handcrafted features extracted from sinus regions. Possible features include intensity statistics, texture descriptors, volume measurements, or shape features. However, this requires manually designing features for each anatomical target. In this project, the abnormalities are region-dependent, multi-label, and sometimes indirectly represented because the dataset does not provide direct septum or turbinate masks. Therefore, handcrafted features alone are unlikely to capture the full appearance variation needed for reliable prediction.

A direct full-volume deep-learning classifier is another possible baseline. In this approach, the complete CT volume or selected slices are passed to a classifier without anatomical localization. Although this is simpler, it is less suitable for this project because the target labels are connected to specific anatomical regions. A full-volume classifier may learn irrelevant correlations or focus on dominant structures while missing rare labels such as maxillary abnormalities.

For these reasons, the proposed method uses a segmentation-assisted pipeline. SwinUNETR first localizes the available sinonasal structures, then anatomy-guided ROIs are extracted and passed to a multi-label classifier. This design is more appropriate for the medical nature of the problem because it connects each prediction to a clinically meaningful region while reducing irrelevant CT information before classification.

4.2 Dataset Description and Annotation Characteristics

The project uses CT volumes and segmentation labels from the NasalSeg dataset together with the provided clinical annotation workbook. The segmentation task uses 130 CT cases with five annotated anatomical structures: right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. These anatomical labels were used for localization and ROI extraction, not as direct abnormality labels.

For classification, the annotation workbook was converted into a model-ready multi-label table. Each selected clinical finding was mapped to an independent binary target, because one CT case may contain several abnormalities at the same time. The final diagnostic targets were nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy.

The annotation workbook originally contained more detailed textual or categorical descriptions for each anatomical region. However, these detailed values **were not used** as separate classes because the dataset size was limited and the positive cases within each region were already small. For example, even if an anatomical region contained around 60 positive cases, dividing these cases into many abnormality subtypes would create extremely small and imbalanced subclasses. This would make supervised learning unstable and would also make evaluation unreliable.

Therefore, the diagnostic annotations were reformulated into binary present/absent labels for each selected anatomical target. This design preserved the clinically relevant question of whether an abnormality exists in each region while avoiding excessive class fragmentation.

The Normal column was not used as a separate sixth output. Instead, a case without any of the selected abnormality labels was represented by an all-zero target vector. This avoided mixing a normal/not-normal decision with the five abnormality-specific outputs and kept the learning problem consistent with multi-label classification.

The workbook also revealed a strong imbalance between labels. Nasal septum abnormality and inferior turbinate hypertrophy had substantially more positive cases than the maxillary abnormality labels, which contained only 22 right maxillary and 23 left maxillary positives. This imbalance influenced the later choice of focal loss, positive class weighting, threshold tuning, and 5-fold out-of-fold evaluation.

A second important characteristic was the mismatch between available segmentation masks and some diagnostic targets. Maxillary labels had direct anatomical masks, but nasal septum and inferior turbinate labels did not. Therefore, the final classifier used contextual ROIs: both nasal cavities for septum-related prediction and the ipsilateral nasal cavity for inferior turbinate hypertrophy.

Table 4.1: Dataset summary

Item	Value
Total matched CT cases	130
Normal cases in annotation workbook	18
Cases with at least one selected abnormal label	107
Final classification task	Five-label multi-label abnormality prediction
Segmentation classes	Background + five anatomical structures
Classification evaluation	Balanced 5-fold out-of-fold evaluation

Table 4.2: Final target labels and positive sample counts

Target label	Positive count	Interpretation
nasal_septum_abnormal	71	Any non-empty septum abnormality annotation mapped to positive.
right_maxillary_abnormal	22	Right maxillary sinus abnormality present.
left_maxillary_abnormal	23	Left maxillary sinus abnormality present.
right_inferior_turbinate_hypertrophy	60	Right inferior turbinate hypertrophy present.
left_inferior_turbinate_hypertrophy	60	Left inferior turbinate hypertrophy present.

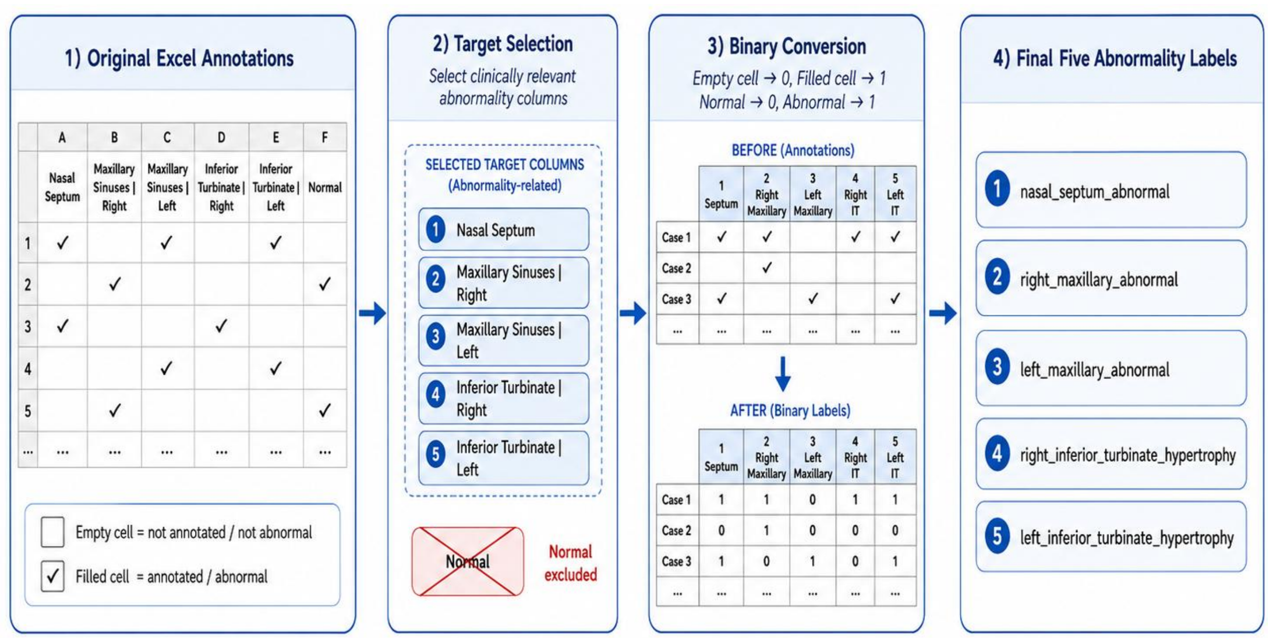

Figure 4.2: Label preparation workflow from the annotation sheet

4.3 Data Preprocessing

Data preprocessing was performed separately for the segmentation and classification stages because each stage uses a different input representation. The segmentation stage works directly on 3D CT volumes, while the classification stage uses cropped anatomy-guided ROIs converted into 2.5D images.

For the segmentation stage, CT volumes were loaded as 3D medical images and processed using MONAI transforms. The main preprocessing operations included orientation handling, spacing normalization, intensity windowing, intensity scaling, and conversion to tensors. Spacing normalization was used to reduce variation in voxel size between CT cases. The selected input spacing was 1.0 × 1.0 × 1.5 mm, which provides a balance between anatomical detail and computational memory cost. The HU window was set to -400 to 1000. This range preserves useful sinonasal air, soft-tissue, and bony information while reducing the effect of extreme intensity values outside the relevant anatomical range.

For the classification stage, preprocessing started after the CT volume and its corresponding segmentation mask were matched. The segmentation mask was used to define the anatomical ROI for each target label. Each extracted ROI contained the CT intensity crop and anatomical mask context. The purpose of this design was to provide the classifier with both appearance information and localization information. The CT crop represents the visual pattern of the tissue, while the mask context informs the model about the anatomical region being analyzed.

After ROI extraction, each crop was resized to a fixed spatial size to ensure that all samples had a consistent input format. This is necessary because sinus anatomy differs between patients and because the bounding boxes extracted from masks may have different sizes. The resized ROI was then converted into a 2.5D image using representative axial, coronal, and sagittal views. These three views were stacked as channels to create an input compatible with pretrained two-dimensional classification backbones.

This preprocessing strategy was chosen to reduce unnecessary anatomical variation and to focus the classifier on clinically relevant regions. It also reduced memory requirements compared with training a full 3D classifier on entire CT volumes, which is important because the dataset contains only 130 matched CT cases.

4.4 Data Augmentation

Data augmentation was applied only to the training data in order to improve robustness without contaminating validation or test evaluation. Validation and test samples were processed using deterministic transforms only. This ensured that reported metrics reflected model generalization rather than random augmentation effects.

For segmentation, augmentation was applied to the 3D CT volumes and masks using spatial and intensity transformations. Spatial augmentation included random cropping, flipping, and rotation. These operations help the model become more robust to small differences in patient positioning and anatomical shape. Intensity augmentation included Gaussian noise and intensity shift or scale transformations, which help the model handle variation in CT appearance and acquisition conditions.

For classification, augmentation was applied to the 2.5D ROI images rather than to the full CT volume. The goal was to improve robustness of the ROI classifier while preserving the clinical label. Light augmentation was preferred because strong geometric transformations can distort anatomical orientation, which is important in sinus CT. The final augmentation strategy avoided creating saved duplicate ROI images because earlier experiments showed that saved augmented copies mainly produced near-duplicate versions of the same patients. Instead, light on-the-fly augmentation was used during training.

This augmentation strategy reflects the medical nature of the problem. In natural image classification, aggressive augmentation can often be useful, but in medical CT analysis, excessive rotation, cropping, or distortion can remove clinically meaningful spatial relationships. Therefore, augmentation was used conservatively to improve generalization while preserving anatomical validity.

4.5 Data Partitioning and 5-Fold Evaluation

The segmentation notebook used an executed split of 91 training cases, 19 validation cases, and 20 test cases. The classification notebook used balanced 5-fold out-of-fold evaluation across the 130 matched cases. This setup ensured that each case was evaluated by a model that did not train on it. The 5-fold design was selected because the dataset is limited and a single train/test split could produce unstable results, especially for rare labels such as maxillary abnormalities.

Table 4.3: Data partitioning strategy used in this project

Stage	Partitioning	Purpose
Segmentation	91 train / 19 validation / 20 test	Train SwinUNETR and evaluate anatomical segmentation on held-out cases.
Classification	Balanced 5-fold OOF	Compare classifiers while using every case once as validation.
Threshold tuning	OOF predictions	Select label-specific probability thresholds for final metrics.

4.6 SwinUNETR Segmentation Stage

The segmentation stage uses SwinUNETR, a 3D transformer-based U-shaped architecture. The model is trained to predict six classes: background, right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. The training objective is DiceCE loss, which combines overlap-based Dice loss and cross-entropy loss. This is suitable for segmentation because it encourages both voxel-level class prediction and mask overlap.

Table 4.4: SwinUNETR segmentation training configuration

Parameter	Value
Architecture	SwinUNETR
Framework	MONAI / PyTorch
Input spacing	1.0 × 1.0 × 1.5
ROI size	96 × 96 × 32
HU window	-400 to 1000
Loss function	DiceCE loss
Optimizer	AdamW
Validation method	Sliding-window inference
Executed split	91 train / 19 validation / 20 test cases

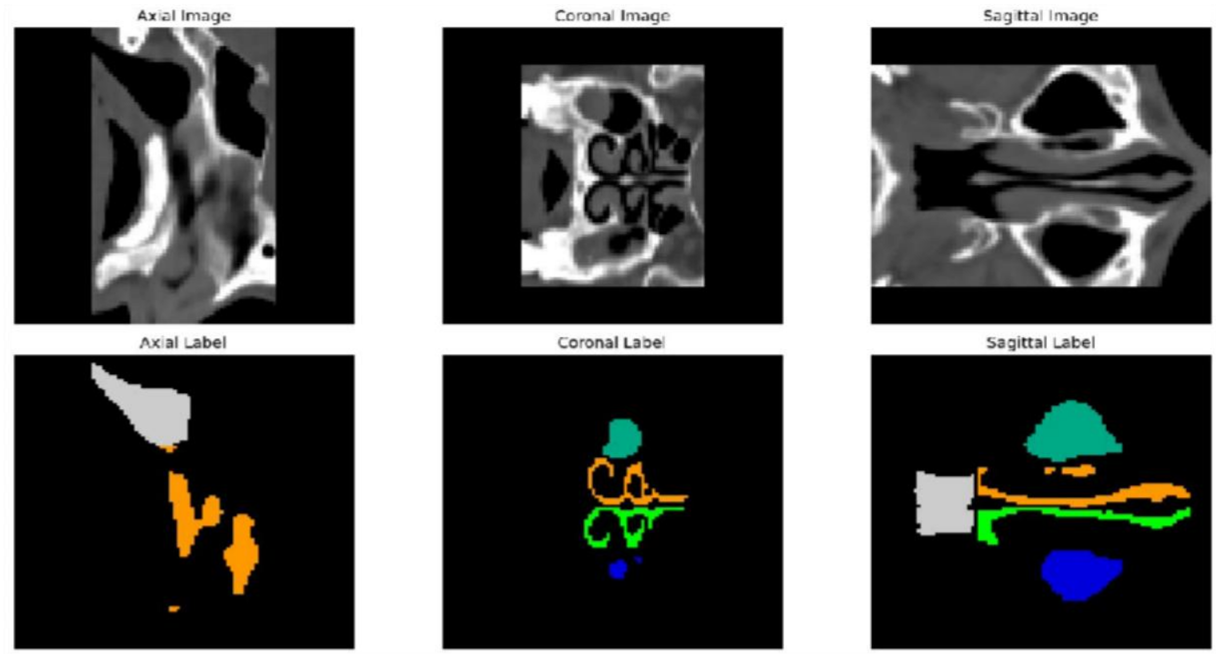

Figure 4.3: Augmentation sanity-check visualization

4.7 ROI Extraction Strategy

The final classification model uses five stable contextual ROIs. The right and left maxillary labels use the corresponding maxillary sinus masks directly. The nasal septum label uses a contextual ROI derived from both nasal cavities with padding. This design follows the anatomical fact that the nasal septum lies between the two nasal cavities, while avoiding brittle pseudo-septum masks. The inferior turbinate labels use the ipsilateral nasal cavity ROIs because direct turbinate masks are not available in the dataset.

Table 4.5: ROI-to-label mapping used in the final classification system

Target label	ROI name	Segmentation labels used	Reasoning
right_maxillary_abnormal	right_maxillary_roi	Right maxillary sinus	Direct anatomical match.
left_maxillary_abnormal	left_maxillary_roi	Left maxillary sinus	Direct anatomical match.

nasal_septum_abnormal	nasal_septum_region_roi	Right + left nasal cavities	Septum lies between both nasal cavities; contextual ROI is more stable.
right_inferior_turbinate_abnormal	right_nasal_cavity_roi	Right nasal cavity	Inferior turbinate is located inside the nasal cavity; no direct turbinate mask available.
left_inferior_turbinate_abnormal	left_nasal_cavity_roi	Left nasal cavity	Inferior turbinate is located inside the nasal cavity; no direct turbinate mask available.

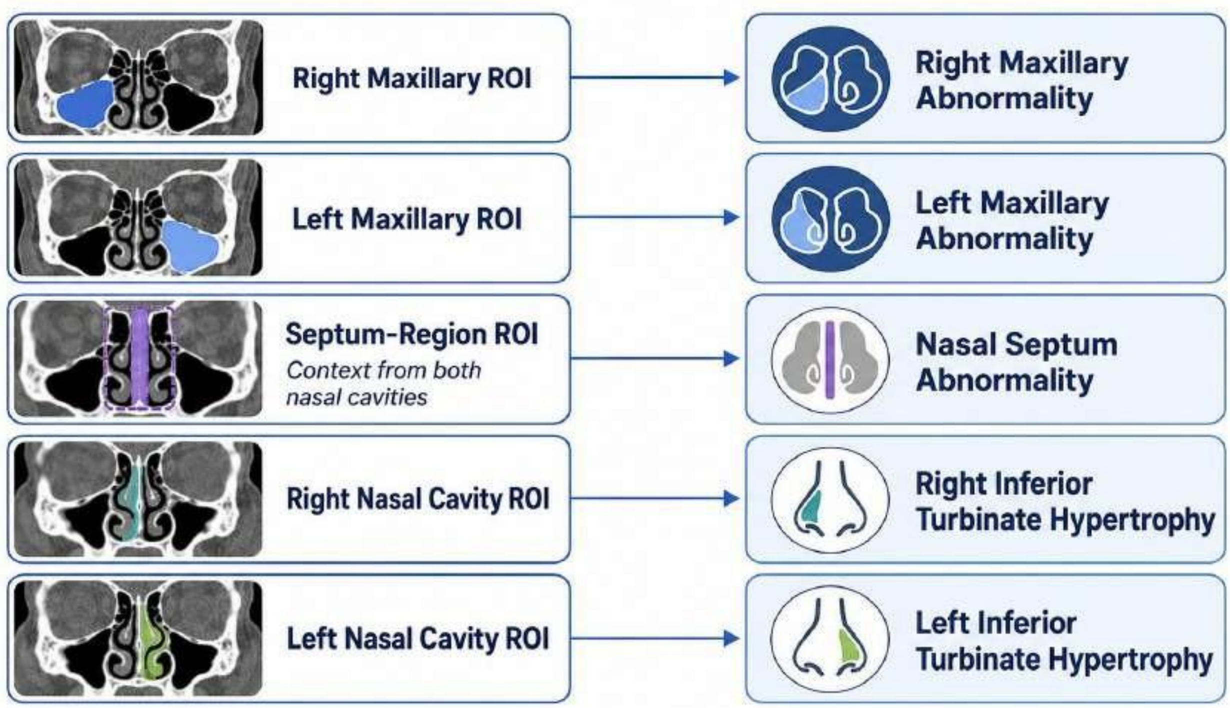

Figure 4.4: Final ROI definitions used in the classification stage

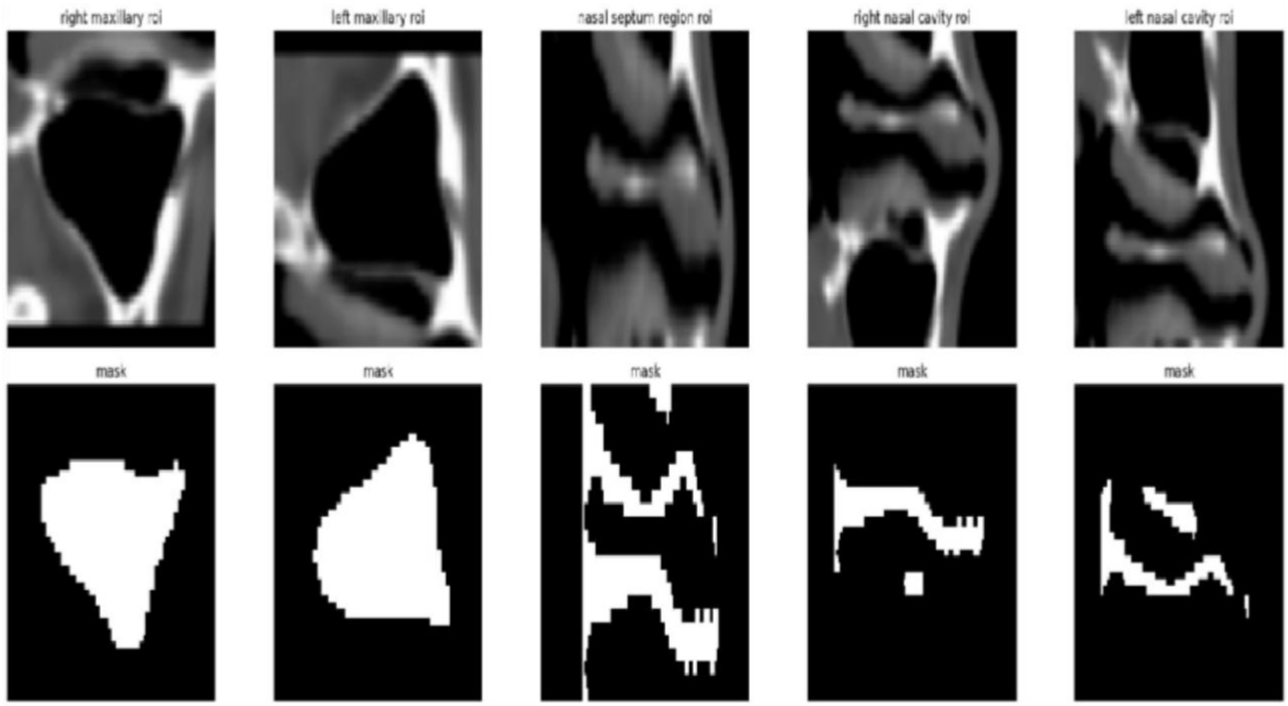

Figure 4.5: Example ROI extraction results from the classification notebook

4.8 2.5D ROI Generation

Each 3D ROI was converted into a 2.5D representation by taking representative axial, coronal, and sagittal views. These three views were stacked as a three-channel image. This design preserved partial volumetric context while allowing the use of pretrained 2D image classification backbones. It also reduced the memory cost compared with fully 3D classifiers, which is important for a dataset of 130 cases.

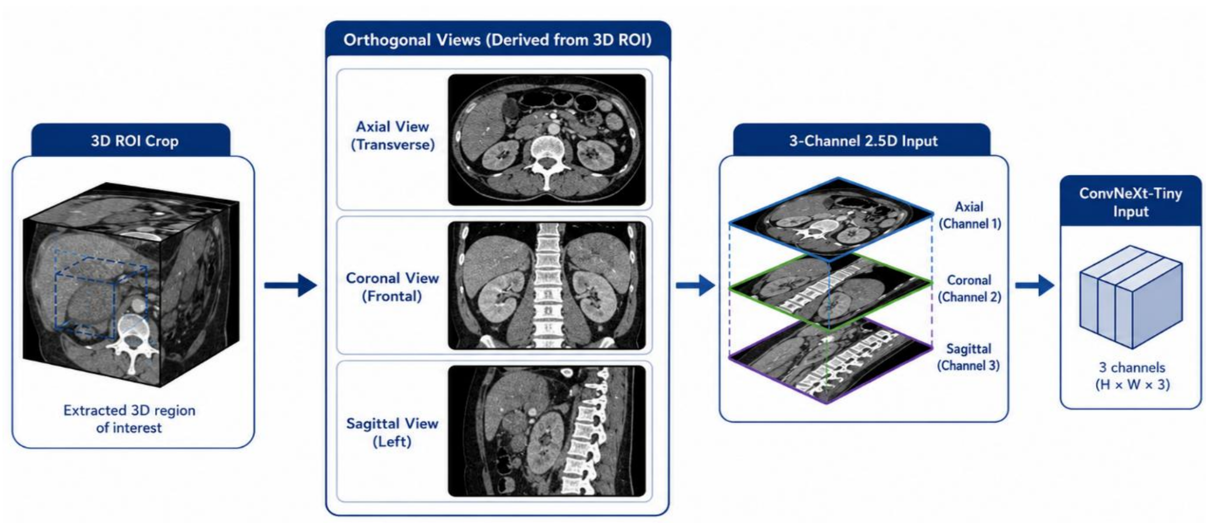

Figure 4.6: 2.5D ROI generation from axial, coronal, and sagittal views

4.9 Important Exploratory Experiments

The classification stage went through several major experimental phases before reaching the current ConvNeXt-Tiny result. This section documents the important development path because the final system was not obtained from a single direct experiment. The project first established anatomical segmentation, then tested ROI-based 3D classification, then moved toward 2.5D transfer learning after several limitations became clear.

The first foundation was the SwinUNETR segmentation model. It segmented the available NasalSeg anatomical structures and provided the masks used to crop regions of interest. This changed the classification task from blind whole-volume prediction into anatomy-guided prediction, where each label could be linked to a clinically meaningful region.

After segmentation, the diagnosis problem was reformulated as multi-label classification. This was necessary because a single sinus CT case can contain septal abnormality, turbinate hypertrophy, and maxillary sinus abnormality at the same time. The model therefore used independent sigmoid outputs rather than a single softmax class.

The first diagnostic models used segmentation-guided 3D ROI crops with a ResNet-style classifier. Early experiments included seven abnormality labels, but the middle turbinate labels had too few positive cases and produced unstable zero or near-zero F1 scores. Therefore, the target set was reduced to the five main labels used in the final system.

The 3D ROI classifier was then improved using two-channel inputs. Each ROI contained both the CT intensity crop and the corresponding mask crop, allowing the model to learn image appearance and anatomical location together. Separate label-specific heads were also more stable than shared bilateral heads, because shared turbinate heads tended to increase false positives.

Several imbalance and augmentation strategies were tested. Raw positive class weighting was too aggressive, so softened square-root positive weights and capped weights were explored. Light on-the-fly CT intensity augmentation was useful, while saved augmented ROI copies were rejected because they mostly created near-duplicate versions of the same patients rather than true new clinical diversity.

Threshold tuning also became necessary. A fixed threshold of 0.5 was not reliable for all labels because each abnormality had a different prevalence and probability distribution. Hybrid and validation-based thresholds improved some fixed-split results, but the results remained sensitive to the split, which motivated the move toward 5-fold out-of-fold evaluation.

The best fixed-split ROI-based 3D ResNet model achieved an approximate macro F1 around 0.63, but later cross-validation showed that this was likely optimistic. In the more honest 5-fold evaluation, the ROI 3D ResNet approach reached an out-of-fold macro F1 of approximately 0.588, confirming that the custom 3D classifier had moderate capability but also a clear performance ceiling.

Additional diagnostic experiments were useful even when they were not selected. Handcrafted ROI statistics, tighter pseudo-septum crops, bilateral context, shared heads, signed and absolute difference features, and ground-truth mask extraction were all examined. These experiments showed that segmentation quality was helpful but not the only bottleneck; the main limitations were small sample size, rare positive labels, indirect septum and turbinate targets, and threshold instability.

The major improvement came from replacing the custom 3D encoder with a 2.5D transfer-learning classifier. Each 3D ROI was converted into axial, coronal, and sagittal views, enabling the use of pretrained image backbones. ConvNeXt-Tiny was the strongest direction because it improved feature extraction while remaining suitable for the limited dataset size.

In the later focused experiments, ConvNeXt-Tiny with focal loss gamma equal to 1, positive class weighting, light augmentation, strict early stopping, ReduceLROnPlateau, and threshold tuning achieved the strongest fold-level macro F1. Swin-T was retained as an important transformer comparison experiment, but it did not outperform the final ConvNeXt-Tiny configuration.

Overall, the experimental history shows that the final pipeline was shaped by systematic failure analysis as much as by successful results. The project moved from segmentation, to ROI-based 3D diagnosis, to honest 5-fold evaluation, and finally to 2.5D ConvNeXt-Tiny transfer learning because each stage exposed a specific limitation of the previous design.

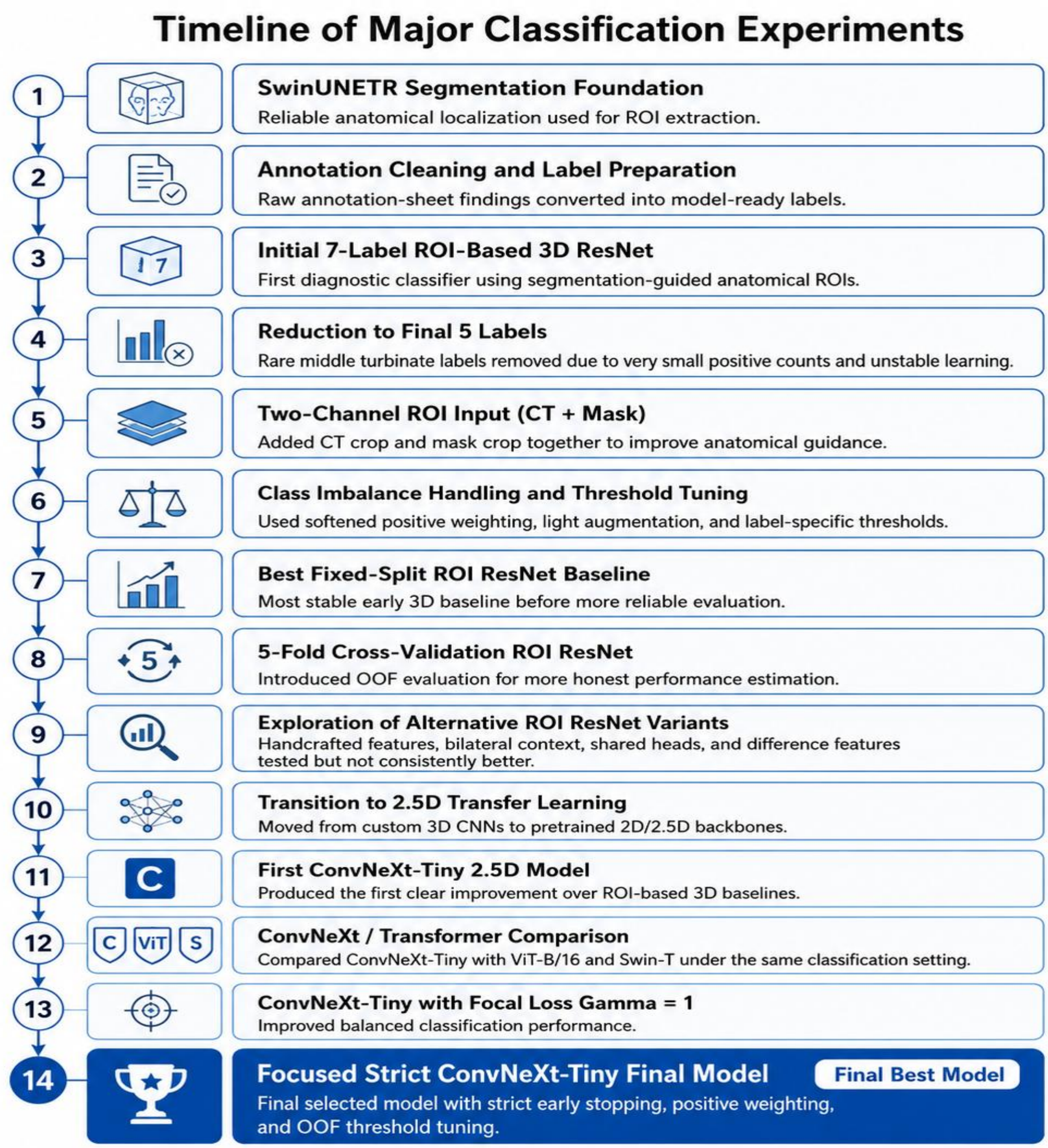

Figure 4.7: Timeline of major classification experiments

Table 4.6: Timeline of major exploratory experiments and outcomes

Stage	Experiment / Idea	Main observation	Decision
1	SwinUNETR segmentation foundation	Mean Dice about 0.87; masks localized maxillary sinuses, nasal cavities, and nasopharynx.	Kept as anatomical localization stage.
2	Annotation cleaning and multi-label formulation	Workbook findings were converted into five independent abnormality labels; normality was represented by all-zero targets.	Adopted for all diagnostic experiments.
3	Initial seven-label ROI 3D ResNet	First full segmentation-guided diagnostic classifier; middle turbinate labels failed because of very small positive counts.	Reduced task to five labels.
4	Five-label 3D ROI classifier with CT + mask input	Two-channel ROI input and separate label heads gave the most stable 3D ResNet direction.	Kept as main 3D baseline.
5	Imbalance handling and augmentation	Softened positive weights, light CT augmentation, and label-specific thresholds reduced some instability.	Retained only stable components.
6	Best fixed-split 3D ROI ResNet	Approximate macro F1 around 0.63, but performance depended strongly on the selected split and thresholds.	Used as fixed-split baseline only.
7	Five-fold 3D ROI ResNet evaluation	OOF macro F1 about 0.588; showed that fixed-split results were optimistic.	Adopted 5-fold OOF evaluation.
8	Alternative ROI and head designs	Handcrafted features, tighter septum ROIs, bilateral context, shared heads, and difference features did not consistently improve results.	Rejected in favor of simpler stable design.
9	Ground-truth mask experiment	Oracle masks improved some ranking behavior but did not clearly improve thresholded OOF F1.	Confirmed labels/data were main bottleneck.
10	2.5D transfer-learning transition	Converted ROIs into axial, coronal, and sagittal views and used pretrained backbones.	Adopted for final classifiers.
11	ConvNeXt-Tiny focal-loss experiments	ConvNeXt-Tiny with focal loss gamma=1 produced the strongest fold-level macro F1 among tested backbones.	Selected as final direction.
12	Swin-T transformer comparison	Swin-T was competitive and useful as a transformer baseline, but remained below ConvNeXt-Tiny.	Kept as comparison, not final model.

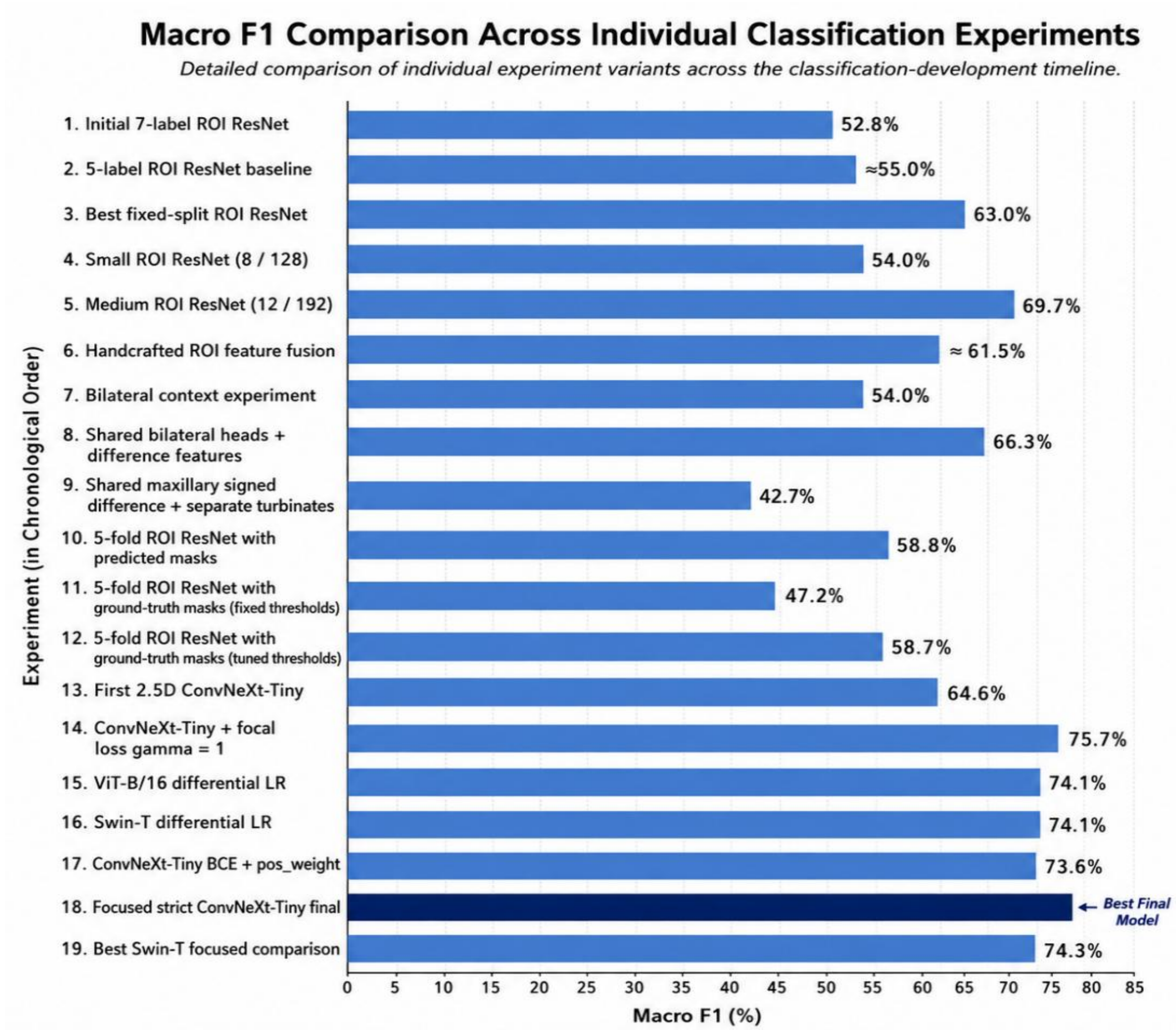

Figure 4.8: Macro F1 comparison across experiments.

4.10 Final Selected Classification Model

The final selected classification model is ConvNeXt-Tiny. It was selected after comparing it with Swin-T under the same segmentation-guided 2.5D ROI classification setting. Swin-T was kept as an important transformer-based comparison model, but ConvNeXt-Tiny achieved stronger fold-level and out-of-fold performance in the final experiments.

The input to the classifier is a 2.5D ROI image generated from a specific anatomical crop. Each ROI is represented using axial, coronal, and sagittal views stacked as three channels. The model therefore receives localized anatomical information rather than the full CT volume. This is important because the target labels are region-dependent and because the dataset size is limited.

ConvNeXt-Tiny was used as a pretrained feature-extraction backbone. The original classification layer was replaced with a project-specific output layer containing five units, one for each abnormality label: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. Because the task is multi-label, the model uses independent sigmoid outputs instead of a softmax layer. This allows more than one abnormality to be predicted for the same CT case.

The final training objective was focal loss with gamma equal to 1 and positive class weighting. Focal loss was used because the dataset is imbalanced and because some labels, especially maxillary abnormalities, have few positive examples. Positive class weighting further helps reduce the bias toward negative predictions. The combination of focal loss and weighting was selected because earlier experiments showed that naive weighting alone could be too aggressive, while the final softened setup produced more stable behavior.

The final training pipeline included light augmentation, strict early stopping, learning-rate reduction on plateau, and 5-fold out-of-fold evaluation. Early stopping was used to reduce overfitting, which is a serious risk with only 130 matched CT cases. Threshold tuning was performed on the OOF predictions to choose label-specific probability cutoffs instead of using a fixed threshold of 0.5 for all labels.

ConvNeXt-Tiny was selected over Swin-T for two main reasons. First, it achieved better final metrics in the project experiments. Second, its convolutional inductive bias is suitable for localized ROI images, especially in limited-data medical imaging. The final model therefore represents a balance between modern transfer learning, anatomical localization, and practical robustness on a small multi-label sinus CT dataset.

4.11 Implementation Environment

The experiments were implemented in a notebook-based Python environment. The segmentation pipeline used PyTorch and MONAI for 3D medical image processing, transforms, training, and sliding-window inference. The classification pipeline used PyTorch, TorchVision, pandas, NumPy, scikit-learn, and Matplotlib. Training was performed in a Kaggle GPU environment.

The final notebooks produced checkpoint files, ROI caches, OOF prediction tables, threshold tables, metrics CSV files, and plots used for analysis.

4.12 Chapter Summary and Research Contribution

The methodology combines 3D anatomical segmentation with ROI-based 2.5D multi-label classification. The major methodological decision is to avoid using the full CT volume as an unlocalized classifier input and instead use segmentation-guided anatomical crops. This makes the pipeline more interpretable and allows each output label to be associated with a clinically meaningful region. The final selected classifier is ConvNeXt-Tiny, while Swin-T is documented as a previous transformer comparison experiment rather than a second final model.

Chapter Five: Research Results

5.1 Introduction

This chapter presents the experimental results of the proposed sinus CT analysis system. The results are reported in two main parts: segmentation results and classification results. Segmentation results evaluate the SwinUNETR model as the anatomical localization stage. Classification results evaluate the final multi-label abnormality prediction stage and compare ConvNeXt-Tiny with Swin-T.

5.2 Tools and Evaluation Methods

The experiments were implemented using Python, PyTorch, MONAI, Torchvision, NumPy, pandas, scikit-learn, and Matplotlib. Training was performed in a Kaggle GPU environment. The segmentation model was evaluated with Dice, IoU, precision, recall, inference time, and throughput. The classification models were evaluated with macro accuracy, macro precision, macro recall, macro specificity, macro balanced accuracy, macro F1, macro AUC, macro average precision, overall label accuracy, and exact match accuracy.

5.2.1 Experimental Environment

The segmentation and classification experiments were developed as notebook-based workflows. This allowed step-by-step inspection of data loading, preprocessing, ROI extraction, model training, metric computation, and figure generation. The final artifacts included trained checkpoints, ROI cache files, fold summaries, out-of-fold predictions, tuned thresholds, per-label metrics, and plots.

5.2.2 Evaluation Metrics Used in the Final Experiments

Segmentation was evaluated using mean Dice, mean IoU, precision, recall, inference time, and throughput. Classification was evaluated using label-wise and macro metrics. Because the task is multi-label and imbalanced, macro F1 and out-of-fold metrics were emphasized over simple accuracy. Exact match accuracy was also reported, but it is naturally strict because it requires all five labels of each case to be correct at the same time.

5.3 Segmentation Results

The SwinUNETR model achieved strong segmentation performance on the test set. The mean Dice score was 0.8743 and the mean IoU was 0.8102. The high recall value indicates that the model generally captured most of the reference anatomical regions. The segmentation results are suitable for the intended purpose of guiding ROI extraction in the classification stage.

5.3.1 Quantitative Segmentation Results

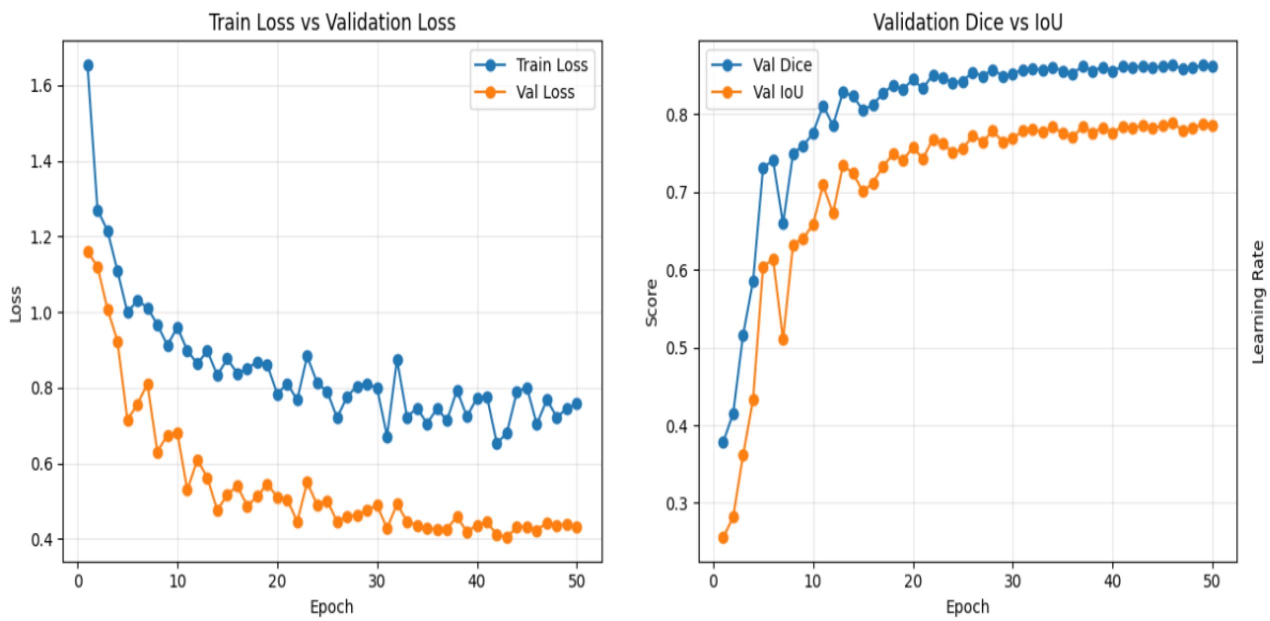

Figure 5.1: SwinUNETR training and validation curves

Table 5.1: Overall segmentation performance on the test set

Metric	Value
Mean Dice	0.8743
Mean IoU	0.8102
Mean Precision	0.8424
Mean Recall	0.9158
Inference time	0.34 s / volume
Throughput	2.90 volumes / s

The segmentation results indicate that the SwinUNETR model was able to provide reliable anatomical localization for the downstream classification stage. The mean Dice score of 0.8743 and mean IoU of 0.8102 show strong overlap between the predicted masks and the reference anatomical labels. The recall value of 0.9158 is particularly important for this project because the segmentation masks are used to extract regions of interest. A high recall means that the model usually captures most of the target anatomical region, reducing the risk that relevant CT information is excluded before classification. Although precision is slightly lower than recall, this is acceptable for the purpose of ROI generation because a slightly larger anatomical crop is less harmful than missing part of the target region.

Table 5.2: Per-class segmentation performance

Class	Dice	IoU	Precision	Recall
Background	0.9896	0.9796	0.9946	0.9846
Right maxillary sinus	0.9107	0.8543	0.8797	0.9476
Left maxillary sinus	0.8697	0.8104	0.8503	0.9053
Right nasal cavity	0.8009	0.7010	0.7440	0.8729
Left nasal cavity	0.8079	0.7138	0.7520	0.8772
Nasopharynx	0.8669	0.8024	0.8338	0.9075

The per-class results show that the maxillary sinuses achieved the strongest non-background segmentation performance, especially the right maxillary sinus with a Dice score of 0.9107. This is useful for the classification stage because right and left maxillary abnormality labels depend directly on the corresponding maxillary sinus ROIs. The nasal cavity classes achieved lower Dice scores than the maxillary sinuses, with Dice values of 0.8009 and 0.8079 for the right and left nasal cavities. This is expected because the nasal cavity has a more complex and irregular shape than the maxillary sinus and contains narrow anatomical spaces. However, these results remained sufficient for contextual ROI extraction, especially for nasal septum and inferior turbinate labels, where the nasal cavity masks are used as anatomical guides rather than final diagnostic outputs.

5.3.2 Qualitative Segmentation Results

In addition to quantitative metrics, qualitative inspection was performed to verify that the predicted masks followed the expected sinonasal anatomy. This step is important because high global segmentation scores may hide local errors that affect ROI extraction. The qualitative examples were inspected in axial, coronal, and sagittal views to ensure that the predicted regions corresponded to the right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx.

Figure 5.2 shows a representative segmentation result on a test case. The predicted mask generally follows the anatomical boundaries of the sinus and nasal cavity regions. The maxillary sinus regions are clearly localized, which supports their use as direct ROIs for right and left maxillary abnormality classification. The nasal cavity regions also provide useful contextual localization for nasal septum and inferior turbinate-related classification, even though the septum and turbinates are not directly annotated as segmentation classes in the NasalSeg dataset.

Some segmentation difficulty can still occur around narrow nasal passages, thin bony boundaries, and irregular mucosal regions. These areas are more challenging because small voxel-level shifts may reduce Dice or IoU even when the predicted region remains clinically useful for ROI extraction. Therefore, the segmentation output should be interpreted as an anatomical localization tool rather than a final clinical segmentation system. For the purpose of this project, the qualitative results confirm that SwinUNETR provides sufficiently stable anatomical guidance for the downstream ROI-based classification stage.

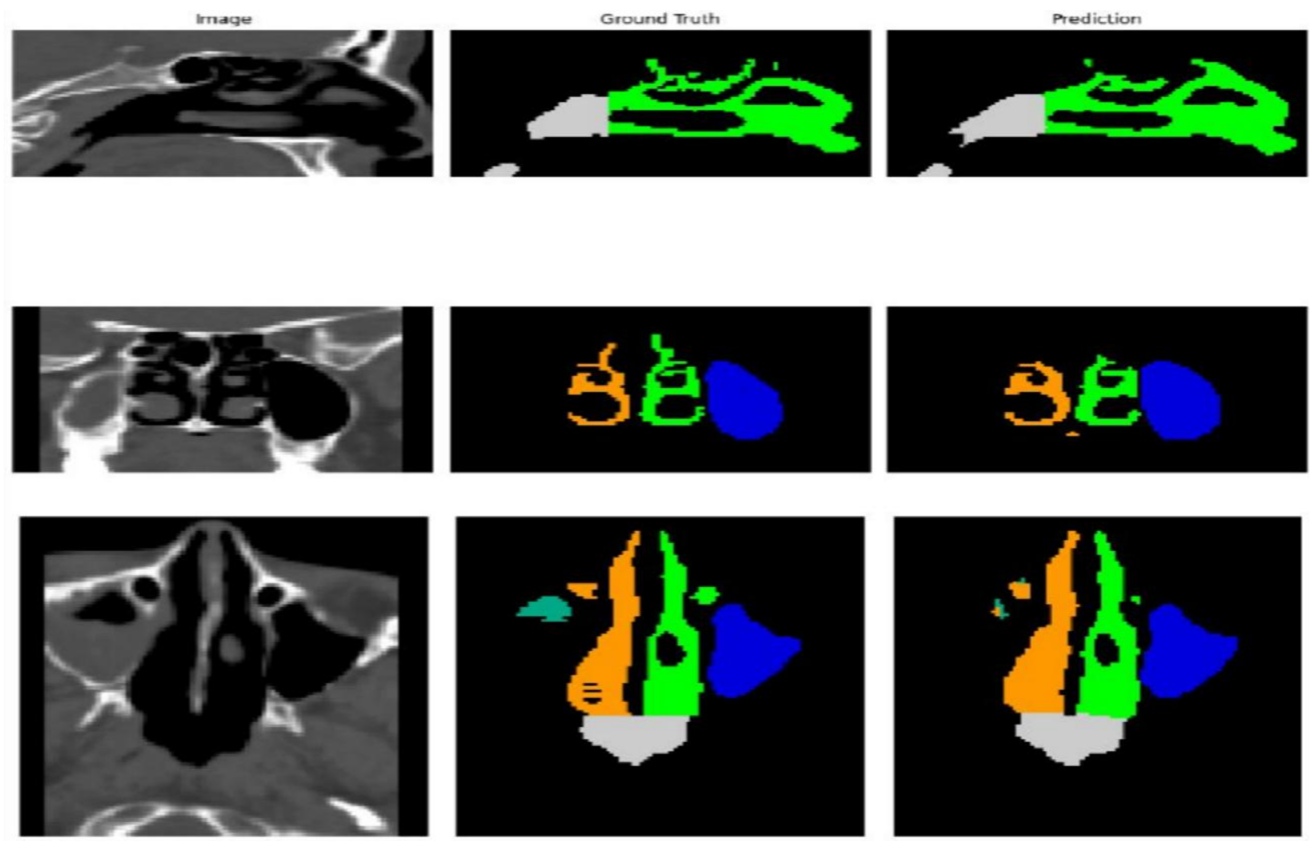

Figure 5.2: Example segmentation prediction on a test case

5.4 Classification Results

The classification stage evaluates the ability of the proposed system to predict five abnormality labels from segmentation-guided 2.5D ROIs. Unlike a single-label classification task, this problem is multi-label because one CT case may contain more than one abnormality at the same time. Therefore, the results are interpreted using out-of-fold macro F1, per-label metrics, and threshold-tuned predictions rather than relying only on simple accuracy.

ConvNeXt-Tiny was selected as the final classifier because it achieved the best overall performance among the final evaluated models. Swin-T was retained as a transformer-based comparison model, but it did not outperform ConvNeXt-Tiny. As shown in Table 5.3, ConvNeXt-Tiny achieved a fold-level macro F1 of 0.7849 and an out-of-fold macro F1 of 0.6928, compared with 0.7401 and 0.6565 for Swin-T. ConvNeXt-Tiny also achieved higher overall label accuracy and exact match accuracy. These results support the selection of ConvNeXt-Tiny as the final classification backbone for the proposed sinus CT analysis system.

Table 5.3: Supporting model-selection comparison between ConvNeXt-Tiny and Swin-T

Model	Fold-level macro F1	OOF macro F1	Overall label accuracy	Exact match accuracy
ConvNeXt-Tiny	0.7849	0.6928	0.7877	0.3846
Swin-T	0.7401	0.6565	0.7031	0.1923

5.4.1 Final ConvNeXt-Tiny Results

The final ConvNeXt-Tiny classifier achieved the strongest classification performance in the project. The main final metric is the out-of-fold macro F1 score of 0.6928 because it evaluates each case using a model that did not train on that case. This makes it more reliable than a single train-test split, especially with a limited dataset of 130 CT cases. The model also achieved an overall label accuracy of 0.7877, meaning that most individual label decisions across the five abnormality targets were predicted correctly.

The exact match accuracy was 0.3846. This value is lower than the overall label accuracy because exact match accuracy is very strict in multi-label classification. A case is counted as correct only if all five labels are predicted correctly at the same time. Therefore, a case with four correct labels and one incorrect label is considered fully incorrect under exact match accuracy, even though it may still be clinically partially correct. For this reason, exact match accuracy is reported as a supporting metric, while macro F1 and per-label F1 are more suitable for evaluating the final model.

The training and validation curves in Figures 5.3 and 5.4 show the learning behavior of ConvNeXt-Tiny during 5-fold evaluation. These curves were used to monitor overfitting, select stable checkpoints, and confirm that early stopping helped prevent unnecessary training after validation performance stopped improving.

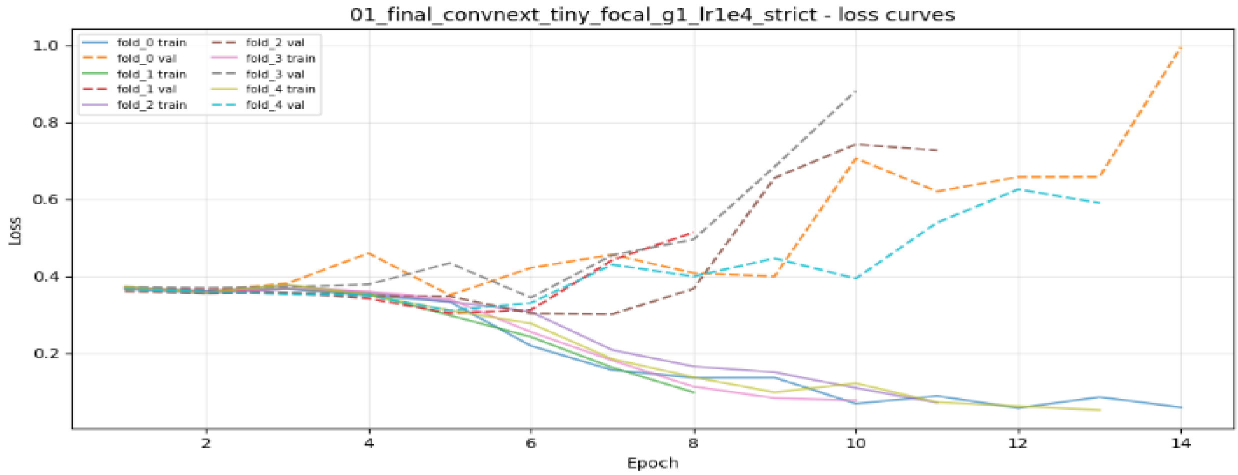

Figure 5.3: ConvNeXt-Tiny loss curves

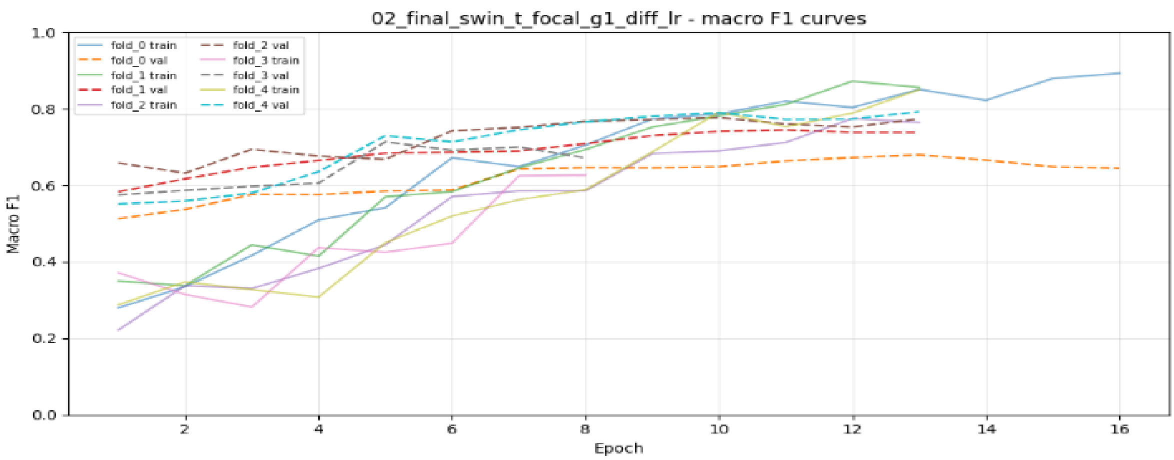

Figure 5.4: ConvNeXt-Tiny macro F1 curves

Table 5.4: OOF per-label metrics for ConvNeXt-Tiny

Label	Threshold	Accuracy	Precision	Recall	F1
left_inferior_turbinate_abnormal	0.49	0.7308	0.7049	0.7167	0.7107
left_maxillary_abnormal	0.42	0.8462	0.5600	0.6087	0.5833
nasal_septum_abnormal	0.44	0.6846	0.6786	0.8028	0.7355
right_inferior_turbinate_abnormal	0.46	0.7769	0.7385	0.8000	0.7680
right_maxillary_abnormal	0.43	0.9000	0.7647	0.5909	0.6667

5.4.2 Per-label Performance Analysis

The per-label results in Table 5.4 show that the final model did not perform equally across all anatomical targets. The strongest result was obtained for right inferior turbinate hypertrophy, with an F1 score of 0.7680. Nasal septum abnormality also achieved a strong F1 score of 0.7355, with high recall of 0.8028. This indicates that the model was able to detect a large portion of positive septum cases, which is important because septum abnormality is one of the most clinically meaningful targets in the project.

The left inferior turbinate label achieved an F1 score of 0.7107, showing relatively balanced precision and recall. This suggests that the nasal cavity ROI provides useful contextual information for turbinate-related prediction, even though the inferior turbinate itself was not directly segmented. The result supports the decision to use ipsilateral nasal cavity ROIs as proxy anatomical regions for turbinate hypertrophy.

The maxillary abnormality labels were more difficult. The right maxillary label achieved an F1 score of 0.6667, while the left maxillary label achieved the lowest F1 score of 0.5833. This behavior is mainly related to the small number of positive maxillary cases in the dataset. The right maxillary label had only 22 positive cases, and the left maxillary label had only 23 positive cases. With such limited positive samples, the model has fewer examples from which to learn the visual patterns of maxillary abnormality. In addition, maxillary abnormalities may co-occur with septal or turbinate abnormalities, making it harder to isolate one label from the others.

Overall, the per-label analysis shows that the proposed system is most stable for septum and inferior turbinate-related abnormalities, while maxillary abnormality prediction remains limited by positive sample count and label co-occurrence. This does not invalidate the ROI-based design, because the maxillary labels use direct maxillary sinus ROIs. Instead, it highlights the need for a larger and more balanced dataset in future work.

5.4.3 Confusion Matrix and Multi-Label Interpretation

The dominant-label confusion matrix in Figure 5.5 provides a compact visual summary of the most confident abnormality predictions. However, it should not be interpreted in the same way as a standard single-label confusion matrix. The classification task in this project is multi-label, meaning that a single CT case may contain more than one abnormality. The dominant-label matrix compresses each case into one dominant true label and one dominant predicted label, which simplifies visualization but loses part of the multi-label information.

This limitation is especially important for cases with co-occurring abnormalities. For example, a patient may have both nasal septum abnormality and maxillary sinus abnormality. If the model correctly predicts both labels but assigns higher confidence to the septum label, the dominant-label matrix may display the case as a maxillary-to-septum confusion, even though the multi-label prediction was partially or mostly correct. Therefore, the matrix may underestimate the model's true usefulness in multi-label cases.

For this reason, Figure 5.5 is used only as a supporting visualization. The final model selection is based mainly on out-of-fold macro F1, per-label F1 scores, and threshold-tuned multi-label metrics. These metrics better reflect the actual diagnostic structure of the problem, where multiple abnormalities may appear in the same CT scan.

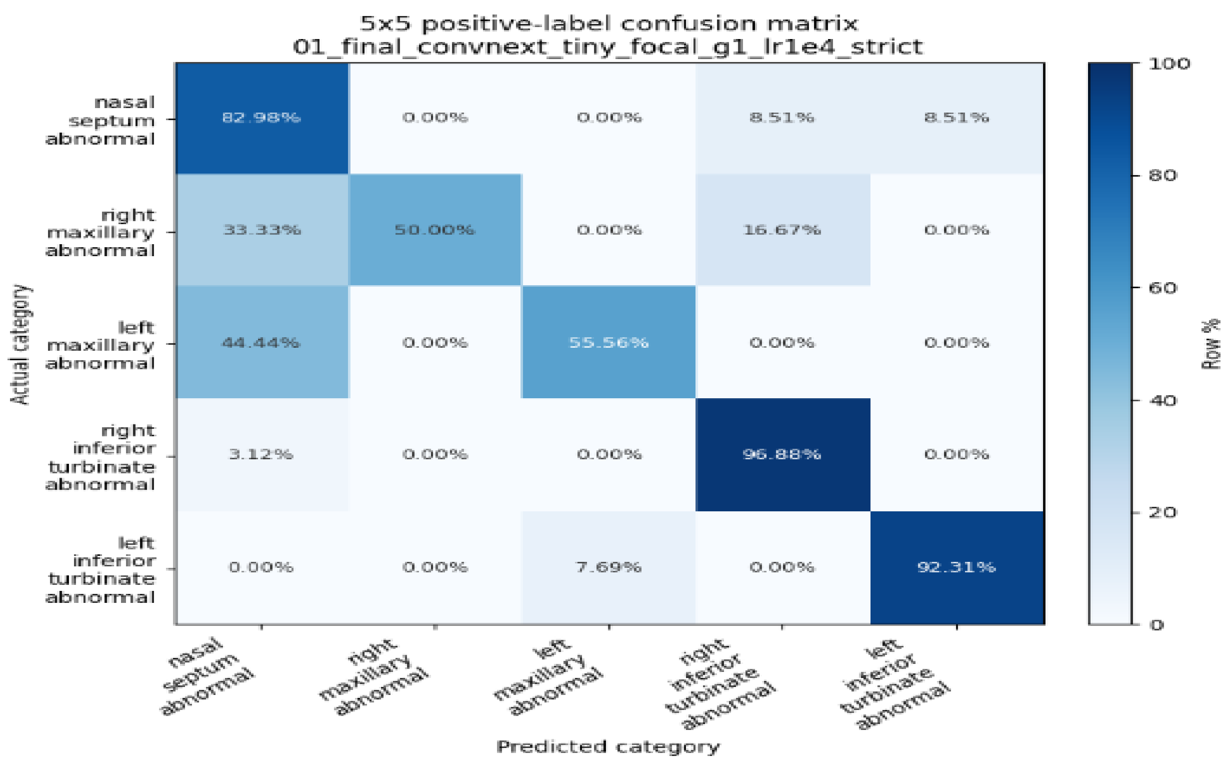

Figure 5.5: ConvNeXt-Tiny 5×5 dominant-label confusion matrix

5.4.4 ROC and Precision-Recall Analysis

ROC and precision-recall curves were used to further analyze the threshold-dependent behavior of the final ConvNeXt-Tiny model. ROC curves show the relationship between sensitivity and false-positive rate across different decision thresholds. They are useful for understanding whether the model can rank positive cases higher than negative cases. However, ROC curves can appear optimistic when the negative class is much larger than the positive class.

Precision-recall curves are especially important in this project because the abnormality labels are imbalanced. They focus directly on positive-class detection by showing the trade-off between precision and recall. This is more informative for rare labels such as right and left maxillary abnormality, where a small number of false positives or false negatives can strongly affect the final F1 score.

The ROC and precision-recall curves showed that ConvNeXt-Tiny learned useful discriminative patterns for several labels, particularly nasal septum abnormality and inferior turbinate hypertrophy. The maxillary labels remained more difficult, which is consistent with the per-label F1 results. This confirms that the main weakness of the final classifier is not only the model architecture, but also the limited number of positive maxillary cases and the co-occurrence of abnormalities within the same CT scan.

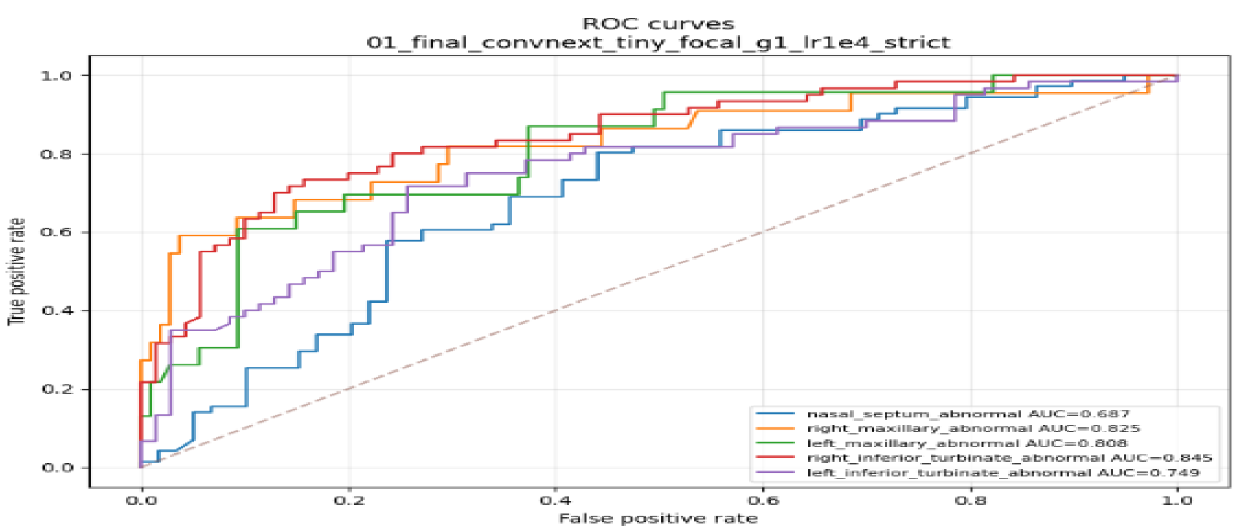

Figure 5.6: ROC curves for the final ConvNeXt-Tiny model

5.5 Comparison with Previous Works

Comparison with previous works must be interpreted carefully because the proposed project does not solve exactly the same task as most related studies. Many previous works focus on segmentation only, while others focus on binary maxillary sinus anomaly classification. In contrast, this project combines anatomical segmentation with five-label multi-label abnormality prediction. Therefore, the comparison is not a direct competition on the same dataset and task, but an analysis of how the proposed system is positioned relative to existing work.

For segmentation, the proposed SwinUNETR model achieved a mean Dice score of 0.8743 and mean IoU of 0.8102 on the test set. This result is consistent with the literature showing that 3D deep-learning models are effective for sinus and paranasal anatomical segmentation. Compared with segmentation-only studies, the main difference is that segmentation in this project is not the final output. Instead, it is used as an anatomical guide for ROI extraction before classification.

For classification, the proposed ConvNeXt-Tiny model achieved an out-of-fold macro F1 score of 0.6928 for five abnormality labels. This task is more complex than binary maxillary anomaly classification because each CT case may contain multiple simultaneous abnormalities. In addition, some target labels do not have direct segmentation masks, especially nasal septum abnormality and inferior turbinate hypertrophy. For this reason, the classification result should be understood as a multi-label research prototype rather than a direct replacement for narrower classification studies.

Table 5.5: Comparison of the proposed model with previous works

Work	Task	Data / scope	Main result	Relation to this project
Zhang et al., NasalSeg [1]	Sinonasal anatomical segmentation	130 CT scans, five structures	Public dataset and benchmark for nasal/paranasal segmentation	Provides the anatomical basis used for ROI extraction
Whangbo et al. [6]	Paranasal sinus segmentation	Sinusitis CT data	Best F1 84.29% on normal test set and 79.32% on abnormal test set	Related segmentation task, but no abnormality classification
Song et al. [10]	Comparison of segmentation architectures	Paranasal sinus CT	Swin UNETR Dice 0.830, Jaccard 0.719	Supports using SwinUNETR-style segmentation for sinus CT
Bhattacharya et al. [7]	Maxillary anomaly classification	Maxillary sinus anomaly task	AUPRC 0.79 with 10% annotated data	Related classification work, but narrower than five-label prediction
Bhattacharya et al. [8]	Maxillary anomaly classification	Maxillary sinus anomaly task	AUROC 0.85 using supervised contrastive	Shows value of deep representation learning, but not

			learning	multi-label sinus CT
Proposed segmentation stage	Anatomical localization	NasalSeg CT cases	Mean Dice 0.8743, mean IoU 0.8102	Provides ROI guidance for classification
Proposed classificat ion stage	Five-label sinus CT abnormality prediction	130 matched CT cases	ConvNeXt-Tiny OOF macro F1 0.6928	Combines segmentation-guided ROIs with multi-label classification

The comparison shows that the proposed system is different from most previous work because it combines two stages: anatomical localization and abnormality prediction. Its segmentation performance is strong enough to support ROI extraction, while the classification results show promising but data-limited multi-label prediction. The main advantage of the proposed system is anatomical interpretability. The main limitation is that the classification stage is trained on only 130 cases with strong label imbalance, especially for the maxillary abnormality labels.

5.6 Discussion of Results

5.6.1 Effect of Segmentation-Guided ROI Extraction

Segmentation-guided ROI extraction was one of the main reasons the final pipeline became interpretable. Instead of asking the classifier to search the whole CT volume, each label was connected to a specific region. This design is especially important in sinus CT because abnormalities are region-dependent. The segmentation stage therefore acts as an anatomical filter before classification.

5.6.2 Effect of Final ROI Design

The final ROI design avoided fragile artificial masks for structures that NasalSeg does not directly annotate. The septum was modeled using both nasal cavities with padding, while inferior turbinate hypertrophy was modeled using the ipsilateral nasal cavity. These contextual ROIs did not eliminate all difficulty, but they were more stable than pseudo-septum or pseudo-turbinate strategies.

5.6.3 Effect of Imbalance Handling

Class imbalance was handled with focal loss and positive class weighting. These choices helped the model pay more attention to difficult and underrepresented labels. However, imbalance was not fully solved, especially for maxillary abnormalities. This shows that algorithmic weighting can reduce imbalance effects but cannot replace the need for more positive cases.

5.6.4 Why ConvNeXt-Tiny Was Selected

During model selection, ConvNeXt-Tiny outperformed the Swin-T comparison in fold-level macro F1, out-of-fold macro F1, overall label accuracy, and exact match accuracy. This result suggests that a modern convolutional backbone was better suited to the available localized ROI data than the transformer comparison model. Therefore, ConvNeXt-Tiny is selected as the final proposed classifier.

The literature review supports this interpretation. ConvNeXt was designed to modernize CNNs while preserving convolutional inductive bias, and several medical-imaging works have adapted ConvNeXt-style designs to limited-data medical settings. Therefore, the choice of ConvNeXt-Tiny is supported both experimentally in this project and conceptually by recent medical-imaging research.

5.6.5 Maxillary Label Instability

The right and left maxillary labels were the most unstable labels. The main reason is the low number of positive cases: 22 for right maxillary abnormality and 23 for left maxillary abnormality. These labels also frequently co-occur with septal or turbinate abnormalities. Co-occurrence makes it harder to isolate the contribution of each label and can make the dominant-label confusion matrix appear worse than the full multi-label prediction actually is.

5.6.6 Limitations of the 5×5 Dominant-Label Confusion Matrix

The 5×5 matrix is useful as a compact visualization, but it is not the main selection metric. It compresses a multi-label case into one dominant true label and one dominant predicted label. This can hide partially correct predictions. For example, a case with both maxillary and septal abnormalities may be counted as confused if the model predicts the correct maxillary label but gives higher confidence to the co-occurring septal label. Therefore, OOF macro F1, per-label F1, and threshold-tuned metrics are more appropriate for evaluating the final model.

5.6.7 Achievements of Research Objectives

The experimental results show that the main technical objectives of the project were achieved to different degrees. The segmentation objective was strongly achieved, as SwinUNETR produced a mean Dice score of 0.8743 and a mean IoU of 0.8102 on the test set. These results demonstrate that the model can localize the available sinonasal anatomical structures with sufficient quality for ROI extraction.

The classification objective was partially achieved. ConvNeXt-Tiny achieved the best performance among the final evaluated classifiers, with an out-of-fold macro F1 score of 0.6928. This shows that the proposed segmentation-guided ROI approach can learn meaningful abnormality patterns from sinus CT data. However, the classification results also reveal important limitations caused by small dataset size, label imbalance, and the absence of direct segmentation masks for the nasal septum and inferior turbinates.

The comparison objective was achieved by evaluating ConvNeXt-Tiny against Swin-T under the same final classification setting. ConvNeXt-Tiny outperformed Swin-T in fold-level macro F1, out-of-fold macro F1, overall label accuracy, and exact match accuracy. Therefore, ConvNeXt-Tiny was selected as the final classification model.

Overall, the project successfully demonstrates a complete research prototype for segmentation-assisted sinus CT abnormality classification. The system is not ready for clinical deployment, but it provides a clear experimental foundation for future improvements, especially through larger datasets, direct septum and turbinate annotations, and external validation.

5.7 Chapter Summary

The results show that SwinUNETR provides strong anatomical segmentation and that ConvNeXt-Tiny is the best final classifier among the evaluated models. The main remaining weaknesses are related to label imbalance, multi-label co-occurrence, and lack of direct masks for septum and turbinates. These issues guide the future work described in the final chapter.

Chapter Six: Conclusion

6.1 Introduction

This chapter summarizes the main findings of the project, states the conclusions, identifies the research limitations, and proposes future directions. The project should be viewed as a research prototype for segmentation-guided sinus CT analysis rather than a clinically deployed system.

6.2 Conclusions

This project developed a segmentation-assisted AI system for sinus CT analysis. The proposed pipeline first uses SwinUNETR to segment key sinonasal anatomical structures, then extracts anatomy-guided regions of interest, and finally applies multi-label classification to predict five selected abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy.

The segmentation objective was strongly achieved. SwinUNETR achieved a mean Dice score of 0.8743 and a mean IoU of 0.8102 on the test set, showing that the available NasalSeg structures can be localized with sufficient quality to support ROI extraction. This confirms the usefulness of anatomical segmentation as the first stage of the proposed sinus CT analysis pipeline.

The classification objective was achieved as a research prototype, but with clear limitations. ConvNeXt-Tiny was selected as the final classifier because it achieved the best performance among the final compared models, with an out-of-fold macro F1 score of 0.6928. This result shows that segmentation-guided 2.5D ROIs can support abnormality prediction, but the classification stage remains limited by the small dataset size, label imbalance, and missing direct masks for the nasal septum and inferior turbinates.

The study also confirms that multi-label sinus CT evaluation must be interpreted carefully. A single CT case may contain several co-existing findings, so per-label metrics and out-of-fold macro F1 are more suitable than relying only on a dominant-label confusion matrix. Overall, the project demonstrates a complete medical-AI workflow for segmentation-guided sinus CT abnormality classification, but further validation and larger annotated datasets are required before any clinical use.

6.3 Research Limitations

- The dataset contains only 130 matched CT cases, which limits the generalization of deep learning models.
- The right and left maxillary abnormality labels have low positive counts compared with other labels.
- NasalSeg does not directly segment nasal septum or turbinate structures, requiring contextual ROI approximations.
- The classification task is multi-label, and co-occurrence makes dominant-label visualization incomplete.
- The final model was not externally validated on data from another hospital or scanner.
- The system is a research prototype and should not be considered clinically deployable without further validation.
- The annotation workbook was converted into binary abnormality labels, so the final classifier predicts abnormality presence or absence but does not model severity, subtype, or detailed radiological grading.

6.4 Challenges and Future Work

The main challenge in this project is the limited size and imbalance of the available labeled dataset. Future work should first focus on increasing the number of CT cases, especially cases with right and left maxillary abnormalities. This would help the classifier learn more stable patterns for the rare labels and reduce the effect of label co-occurrence.

A second important direction is improving anatomical annotation. The current dataset provides masks for maxillary sinuses, nasal cavities, and nasopharynx, but it does not provide direct masks for the nasal septum or inferior turbinates. Adding expert annotations for these structures would allow more precise ROI extraction and reduce dependence on contextual proxy regions.

The system should also be validated on external CT data from another scanner, hospital, or acquisition setting. External validation is necessary before judging whether the model generalizes beyond the current dataset. In addition, radiologist review should be used to improve label quality and standardize abnormality definitions.

From a modeling perspective, future work can explore hybrid 2.5D/3D classifiers after collecting more labeled cases. The current 2.5D ConvNeXt-Tiny model was suitable for the limited dataset, but a larger dataset may allow stronger volumetric models. Calibration and uncertainty estimation should also be studied so that the system can express prediction confidence more safely.

Finally, a future clinical interface could display the CT slices, segmentation masks, extracted ROIs, and predicted abnormality labels together. Such an interface would make the system more useful as an educational or decision-support prototype, while keeping the final clinical decision under expert supervision.

Table 6.1: Summary of main findings and future work directions

Finding	Implication	Future direction
SwinUNETR segmentation performed well	Segmentation can guide ROI extraction	Add improved or manual masks for missing structures.
ConvNeXt-Tiny outperformed Swin-T	Modern CNNs were more stable for this dataset	Keep ConvNeXt-Tiny as the final baseline.
Maxillary labels were unstable	Low positive counts and co-occurrence affect learning	Collect more maxillary-positive cases.
Dominant-label matrix is limited	Multi-label correctness may be hidden	Use per-label and case-level multi-label metrics.
Septum/turbinate masks were unavailable	Contextual ROIs are approximations	Annotate septum and turbinate structures directly.
Binary labels simplify clinical findings	Severity and subtype are not modeled	Add radiologist-reviewed severity labels in future datasets

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